

Finite element modelling of complex 3D image data with quantification and analysis

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Abstract

The purpose of this study was to examine how to model aggregated material microstructure and its meshing volumic generation that is provided by any data from 3D tomographic image data. The accurate reconstruction of 3D geometry structures from tomographic images is a powerful method in various application areas of materials science. The resulting mesh can be voxelized or conforming based on volumetric tetrahedral meshing. We investigate this creation depending on improving multiple materials marching cubes algorithm (M3C) with smoothing and remeshing algorithms. Then, a strategy for generating good-quality meshing and its robustness is presented, and this is performed with numerical tests. The novelty of this study is to generate a conforming mesh from complicated topology structures, particularly, when the interfaces of bi-materials are connected. This leads to a reduction in the node count in the generated mesh. The influence of some parameters involved in this algorithm is explored during different levels of meshing. In this work, the numerical homogenization approach from various spherical inclusions in the two-phase system using the algorithm M3C is considered to estimate the effective elastic properties. We created the framework with all the associated information, such as inputs in the format .inp files, to make it possible to run it over the Abaqus solver. Then, the Abaqus model based on the finite element method (FEM) was executed in this case for various material microstructures such as polycrystalline, composite, and fiber. We show the main workflow for providing desired results by visualizing the FEM analysis. We also demonstrate the capabilities of meshing methodology in the solver for these material models. The validation of the local mechanical environment from FEM with loading scenarios is achieved to predict displacements and deformations. Mechanical compression tests are performed to investigate the compressive behavior. Finally, stress-strain curves provided a comparison between simulations and experimental data for materials, and a good agreement is obtained.

Keywords: Image processing; marching cubes algorithm; multi-material; voxel-mesh; finite element method

Introduction

A vast range of mesh generation approaches [1, 2] is directly available. However, most of them have not been investigated with meshing from simultaneous segmentation and labeling 3D data [3, 4]. Meshing from 3D imaging data involves many challenges but also opportunities for showing a practical and suitable described geometry of the computed domain. This domain could have a complicated topology [5–7]. Using a comprehensive approach from imaging data with good properties remains demanding.

There are two principal methods of meshing from 3D imaging data. The first is named the CAD-based approach [8, 9]. It is based on the surface reconstruction framework, which is given by a specific meshing algorithm [10, 11]. Most of these techniques are currently used to follow the classical CAD method. These approaches utilize the scan data to describe the domain's surface followed by the created elements marking the limits of a volume area. They are frequently characterized as time-consuming and designed for robust, complex topologies. The generated mesh could contain blocks with non-conforming grid interfaces. The second is the image-based approach [12, 13]. This approach is more direct as it combines the geometric detection and mesh creation stages in one process, offering a robust and accurate result.

Due to the rapid development of reverse engineering based on modern 3D scanning technologies [14, 15], we will focus on the direct approach to tackling surface reconstruction problems [16, 17]. In the present work, we propose the voxel conversion technique, which provides meshes with tetrahedral brick elements. This approach consists of generating 3D tetrahedral elements throughout the volume of the domain, thus creating the mesh directly with conforming multipart surfaces. This paper will show that the obtained mesh is helpful for guaranteeing accurate results during pre-processing steps. Many applications of voxel-mesh modeling are described in different works, one of which is detailed in [18]. The challenge of this work is to get the suitable microstructure geometry with a smooth interface from an image. In this topical study, an existing surface reconstruction method [19] from Marching Cubes algorithms (M3C) [20, 21] has been proposed and strongly applied for the designated microstructure image. The remeshing algorithm [22] is developed for the non-manifold surface with efficient parallel implementation to transform the mesh faster and more optimally in time.

Image-based modeling has various essential applications [23–25] in material science, medicine, and computer graphics fields. It is a solution for obtaining a practical model with simulation in

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measurements acquired experimentally and stay unfeasible. Other approaches exist in material science, which are good ways to verify these results experimentally.

The material microstructure is often considered to be many elements of voxels, i.e. pixels in the bi-dimensional case [26, 27] in which each element voxel corresponds to a physical phase. This microstructure presented in the form of voxels is created from microcomputed tomography (μ -CT) scans [28–30]. In this situation, element voxels do not offer an ideal way of presenting the phases. Furthermore, scanning steps are necessary to identify them perfectly. This case can be extended to the three-dimensional space when the microstructure does not change [31, 32]. The microstructure could be procured using different geometrical approaches, such as Voronoi diagrams [33, 34].

Some microstructure generation is largely performed via traditional tessellation techniques [35, 36], in which are proposed to overcome the topological restrictions of some materials. The deformation of such materials, which includes martensitic steel (P91), is modeled using the finite element method (FEM). In this case, the microstructure generation takes into account the Voronoi Tessellation (VT) method, which consists of decomposing a domain into different regions with polygonal elements. For crystalline materials, these elements define equiaxed grains with some crystal orientation distribution. FEM and VT methods are widely combined in [37, 38] to investigate lattice deformation and failure evolution at the microscale. The combination of these methods focuses on the crystalline material families to investigate how the morphology reacts, in particular, how to explore the dislocation at some locations, such as block and packet boundaries [39].

Research has been carried out to investigate the representation approaches of engineering objects, and review the recent developments of heterogeneous objects [40]. However, identifying the absolute space from a given microstructure continues to be questionable; even this question could be predictable. The published works [41–43] provide the responses and address significant contributions proposing a performed mesh quality in M3C, with improved smoothing algorithms such as Laplacian smoothing or dihedral angle control. There are other mathematical models in both materials science and geometry. For instance, a new geometric method analyzing elastic and elastic-plastic solids [44, 45] is proposed with involving discrete operators in material and mechanical engineering.

In the present work, the image files are managed well with the consistency of the coordinate system. The 3D images can be stored in numerous binary formats. Two principal formats have been handled. The first is named tiff, a famous classical image format for creating FE mesh. The second is the vti format from the vtk framework [46, 47], which is considered a powerful format for storing scalar, vector, and tensor data. Additionally, reading and writing images from our framework follow the traditional convention, which matches the ImageJ [48] convention when looking at the data. It is necessary to recognize that the input images in these supported formats are viewed in reality with the proper type of finite elements (FE) model, including all the microstructure details. The FE modeling will be run after obtaining high-quality mesh responding to the good criteria. The computed surface reconstruction from a three-dimensional scan [49] responding to the optimality criteria [50] is well viewed in this format.

The outline of this paper is as follows. Section “Motivation” focuses on the work’s motivation by examining some standard techniques to assess the behavior of isosurface extraction. The segmentation procedure in which an image is manipulated with a

rotation approach is formulated briefly in Section “Fiber orientations with segmentation aspect”. Next, Section “Multiple material marching cubes algorithm (M3C)” describes the approved 3D reconstruction surface that our FE model can generate. This proposal is utilized to develop the best discretization approach from the input image to obtain the resulting mesh. Section “Meshing robustness in some material types” deals with configured tests in variables involved in this algorithm, leading to a superior presentation regarding the mesh size. In Section “Spherical inclusion”, we provide an important application of our algorithm in the case of spherical inclusion. Typically, the quality mesh between the voxel and the tetrahedral one is analyzed deeply. Section “Computation chain on Abaqus” focuses on the computation chain under Abaqus solver in which various numerical simulation results are presented. Finally, concluding remarks are summarized in Section “Summary and future work”.

Motivation

The most challenging problem with standard techniques for iso-surface extraction from volumetric datasets is to minimize the number of generated triangles or polygons [51, 52]. An early technique begins with constructing and connecting the surface contours with triangles. Unfortunately, if there is more than one contour of surface inside a slice, problems occur for connecting shapes. Then, ambiguities appear when determining which contours to connect. The Marching Cubes (MC) technique is examined as a classical approach to this problem of extracting isosurfaces from a volumetric dataset. This technique is proposed in the first works [53, 54]. This standard MC technique has some specific imperfections, particularly topological inconsistency [55, 56]. It cannot produce a consistent notion of object connectivity. Even though it is a simple and practical algorithm that is available in the software of many commercial systems, its implementation should be controlled. Overcoming some obstacles consisting of topological ambiguities remains challenging, including those in the form of incoherent output surfaces. To convince the reader, a default evaluation of the proposed approach to datasets related to monkey imaging is reported in Fig. 1. Tetrahedral elements are used here as the element type, which is geometrically versatile and can conform to complex boundaries and anomaly structures. This Figure depicts the influence of the ambiguous topology on input geometry. This algorithm can not guarantee the continuity of this proposed geometry. Geometric adaptability [57–59] is necessary for the background mesh to look like the original image. To verify the improved MC algorithm, we will compare the 3D reconstruction results with the existing input image. The improved MC algorithm ensures the perfect meshed topology structure.

This work aims to effectively resolve a topology ambiguity problem inherent in the MC algorithm to generate an appropriate mesh even for an arbitrary complex geometry. Another objective is to diminish induced cells without excessively expanding the overall computational complexity. Several examples, including material modeling, are proposed to demonstrate the adapted mesh generation scheme. Note that since the topology information utilized in this algorithm is based only on surface information, not volume information, the surface reconstruction is improved to make it suitable. Making this change enhances the topology that has been defined previously.

The proposed method focuses on providing an appropriate and reasonably smooth surface. This refined mesh fits with the surface shape and improves the topological configuration

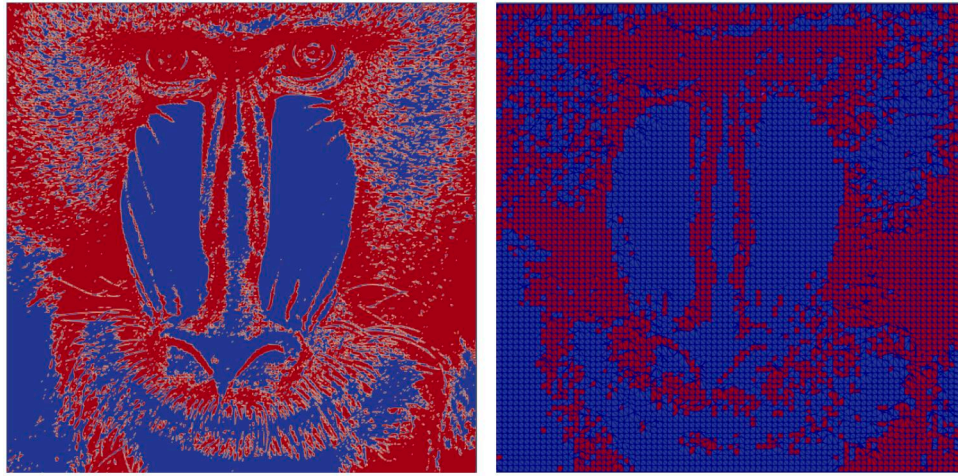


Figure 1. The plan projection of tetrahedral mesh considering ambiguity.

without the void phenomenon [60, 61]. However, for volumetric mesh [62], a new method is implemented to generate the output mesh that can incorporate this surface reconstruction. As an alternative method for structural topology optimization, we will also focus on meshing optimization that is achieved by removing vertices and collapsing edges.

During the mesh processing, the topology of related vertices is verified in order to avoid changing the original one. The criteria in this mesh simplification consists of bringing new vertices near the associated ones. This work proposes an alteration of the MC algorithm for isosurfacing by improving the grid cell representation inside this surface. The goal is to create a mesh representation with the correct topology. This permits us to represent performantly various complex topologies that can occur. The surface, in this case, will be continuous. Further, the topology configuration of the repeating unit cells is also improved since this configuration can significantly influence the mechanical properties. Then, twenty-four topological configurations instead of 15 eliminate the effect of voids. Expanding this number of configurations contributes to refining the topology configuration and meshing the topology structure perfectly. The comparison of the formed triangles validates the accuracy of modifying this configuration. The rapidity of M3C compared to the standard MC is checked. The speed-up has risen by about 35% and increased at the same rate, with ameliorating connectivity and the LUT algorithm, as detailed hereafter.

The aim is to propose a simple but complete method by first abstracting twenty-four expanded configurations from the initial cube size. Several separate triangular patches will be regular for these configurations. The problem with the MC technique is speed. As we will see later, the multi-thread parallel processing method is used to accelerate the new configurations. The speed of surface construction will be improved, and the number of triangles representing the surface will be significantly reduced.

Fiber orientations with segmentation aspect

This section is devoted to reporting the rotational algorithm providing accurate rotation of rigid fibers. Specifically, a micro-mechanical model is described, which is developed using orientation analysis to run tows segmentation algorithms. The model contains fibers that are distributed randomly over a microstructure. This microstructure with complex internal constituents is illustrated in Fig. 2, which is separated into two diagrams. The input 3D

image of the composite material in XYZ coordinate system is presented in the left diagram. Next, X-ray microtomography is utilized to create the μ -CT image shown in the right diagram, presenting the bi-dimensional plane in XY plane. There are three colors shown in this image. The black area indicates air, the dark area represents matrix resin, and the grey area corresponds to tows. Note that a non-uniform distribution of pores is represented by black spheres across the matrix material within the fiber.

The segmentation step would become exigent in the case of two or more zones with the same grey-scale that have identical interfaces and diverge only in texture. This means that the texture is not visible enough signifies a lower resolution of the scan. This prevents us from obtaining an automatic segmentation in high quality. Combining modern segmentation with manual processing allows us to utilize these segmentation functions from our μ -CT data. Simply put, it consists of smoothing and thresholding operations in classical situations. For this purpose, ImageJ [63] is not a practical tool for the segmentation of this image since there are a lot of domains. We refer to the reader to [64–66] for the segmentation tow domain following orientation randomly by an approach. This approach is called finite difference hessian and concerns 2D textile composites. It remains the main step for texture recognition.

Rotations in three-dimensional Euclidean space possess three independent components, as demonstrated by Euler. They are described in terms of Euler angles, orientation matrix, Rodrigues vector, and quaternion. The application of Euler angles is a classical method for describing a rotation with three consecutive rotations around each principle axis. Note here that these angles are computed using the Bunge convention from rotation matrix, as exists in crystallography and texture analysis [67].

Denoting $[0, 2\pi[$ the angle of first rotation around Z-axis, the angle $\Phi \in [0, \pi[$ corresponds to rotation around the new X-axis, and $\phi_2 \in [0, 2\pi[$ the angle of first rotation around the new Z-axis. The overall rotation matrix corresponding to each elementary rotation is determined in the following form:

$$g(\phi_1, \Phi, \phi_2) = \begin{bmatrix} \cos \phi_1 \cos \phi_2 - \sin \phi_1 \sin \phi_1 \cos \Phi & \sin \phi_1 \cos \phi_2 + \cos \phi_1 \sin \phi_2 \cos \Phi & \sin \phi_2 \sin \Phi \\ -\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_1 \cos \Phi & -\sin \phi_1 \sin \phi_2 + \cos \phi_1 \cos \phi_2 \cos \Phi & \cos \phi_2 \sin \Phi \\ \sin \phi_1 \sin \Phi & -\cos \phi_1 \sin \Phi & \cos \Phi \end{bmatrix} \quad (1)$$

The model presented in Fig. 2 recovers three phase behaviors. At one level, there is a tow, a dense carbon fiber studied as a

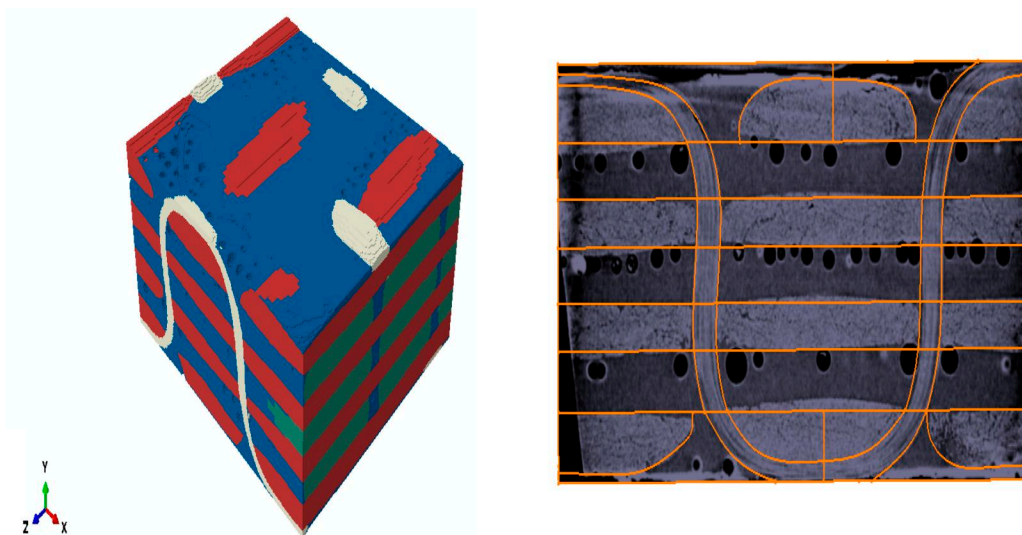


Figure 2. Voxel-mesh model of the textile composites with carbon fiber, accompanied by its 2D slice in XY plane obtained by μ -Computed tomography.

strongly anisotropic material. Its local fiber orientation field needs to be determined, and ellipses could approximate them. These orientation angles are required for the analytical homogenization approaches, which are used to compute the material properties of a tow as mentioned in [68]. These angles are also used in FE modeling for others types of material microstructure.

Multiple material marching cubes algorithm (M3C)

This section addresses a strategy routine based on the MC algorithm to reconstruct multi-material and smoothed surfaces from any given image. For this, we begin this section by briefly evaluating some meshing software. Indeed, there exist commercial softwares providing numerical results for processing of multiply materials images. Some of these softwares like Avizo, BioMesh3D, Cleaver, and MeshLab, have a good interface design for data visualization. For instance, Avizo [69] is able to generate an initial surface mesh, which is based on the general marching cubes algorithm (M3C). However, making smaller meshes with it means that stage optimization is difficult. BioMesh3D [70] is a meshing tool for 3D image data quantification and analysis that is developed for medical applications. Its disadvantage is its slow efficiency, since it generates too many nodes at some particular zones. MeshLab [71] is well-known software for image-based meshing with the best visualization purposes. After this brief review, our original investigation develops the construction of a simple interface mesh with an adequate number of nodes. This interface is smoothly meshed, with a refinement procedure after completing the image preparation step.

Among the MC modifications shown in Figs 3 and 4, our framework is based on the existing algorithm [58]. For instance, the surface topologies of the improved algorithm are consistently expanded in [72] to polyhedral volume topologies. The original algorithm lookup table (LUT) is not used due to the large size. The eight vertices can have up to eight different label materials in a multi-material cube. Each label designs the material ID and is given in the form of a status value. Then, there are 8^8 or 16 777 216 possible combinations for generating all cells. If all possible triangles are taken into account, then the drive's capacity is displayed as 1.2 GB (gigabytes). This quantity means expensive

computation related to this significant operation. For this reason, the LUT algorithm is improved.

The developed volumetric tetrahedral meshing is structured into five stages. The first one is to prepare the input image, and this is optional. It consists of removing small regions and fixing diagonally connected voxels. Next, the initial surface is created using modifications of the multiple material marching cubes algorithm [58]. The third stage is the preliminary remeshing step. It includes extracting boundary edges from the obtained initial surface in order to be smoothed. Then, the fourth step consists of creating the half-edge data structure to follow the remeshing process. The smoothed surface is remeshed with fixed boundary edge nodes during this stage. Finally, TetGen [73] is used and applied to previous surface mesh, providing the resulting tetrahedral mesh. It is important to use this software to generate good quality mesh suitable for our FE Models. Figure 5 summarizes the main workflow, including these different steps. Note that it is difficult to show visually the 3D volumic presentation. However, Fig. 5A–D depict its surface presentation at each stage, in which the volume is subdivided into a number of tetrahedral cells. The fourth and final stages are presented in only one diagram Fig. 5D. The surface mesh presented in Fig. 5C seems to be more refined compared to the one presented in Fig. 5D. However, the volumic cell number presented in Fig. 5D is much higher than its presented in Fig. 5C. This superiority is explained by the impact of the mesh generator TetGen at the final step.

The FE model is based on the tetrahedral mesh that could be differentiated from the corresponding voxel-mesh model result. The voxel representation of geometry is ambiguous, involving an incorrect topology. Additionally, the FE approximation can produce an oscillation at the interface region when using the voxel mesh model. The main disadvantage of this representation is that a low noise level appears during the segmentation step. Other disadvantages are the diagonal voxel connectivity problems and extensive computation resources. In particular, the tetrahedral mesh quality is better than the voxel mesh. Our choice is motivated by the geometry accuracy and the mechanical precision framework. This framework manages the local orientations well and is suitable for the numerical homogenization results.

We now discuss the initial step devoted to manipulating some functionalities for image processing. It is aimed at analyzing

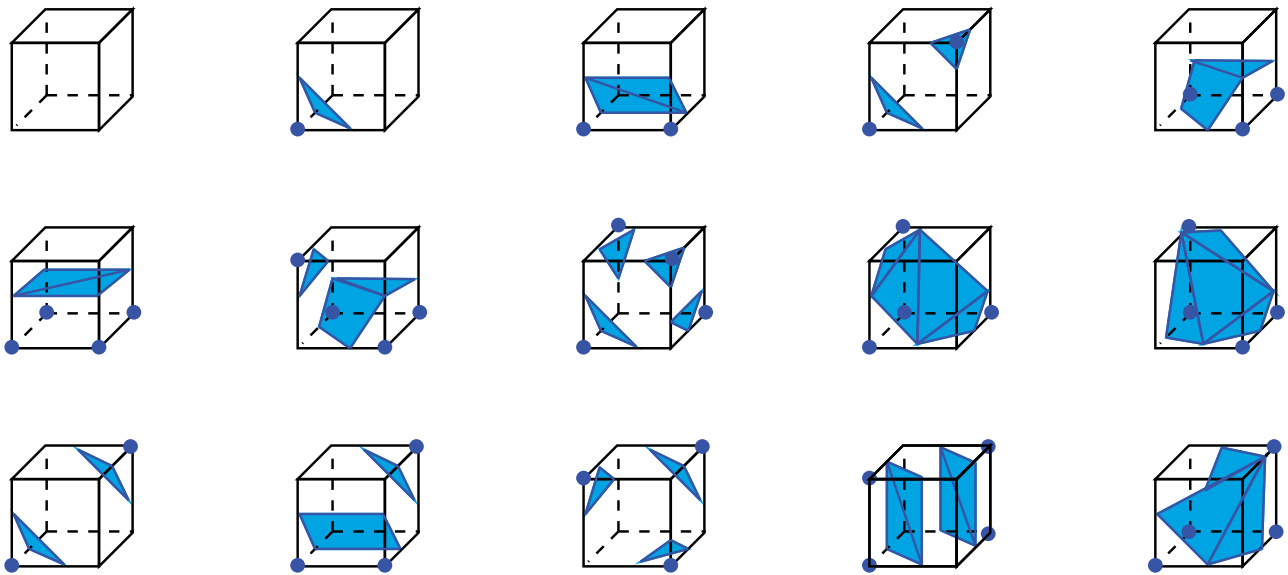


Figure 3. The basic topological configurations of marching cubes algorithm (MC).

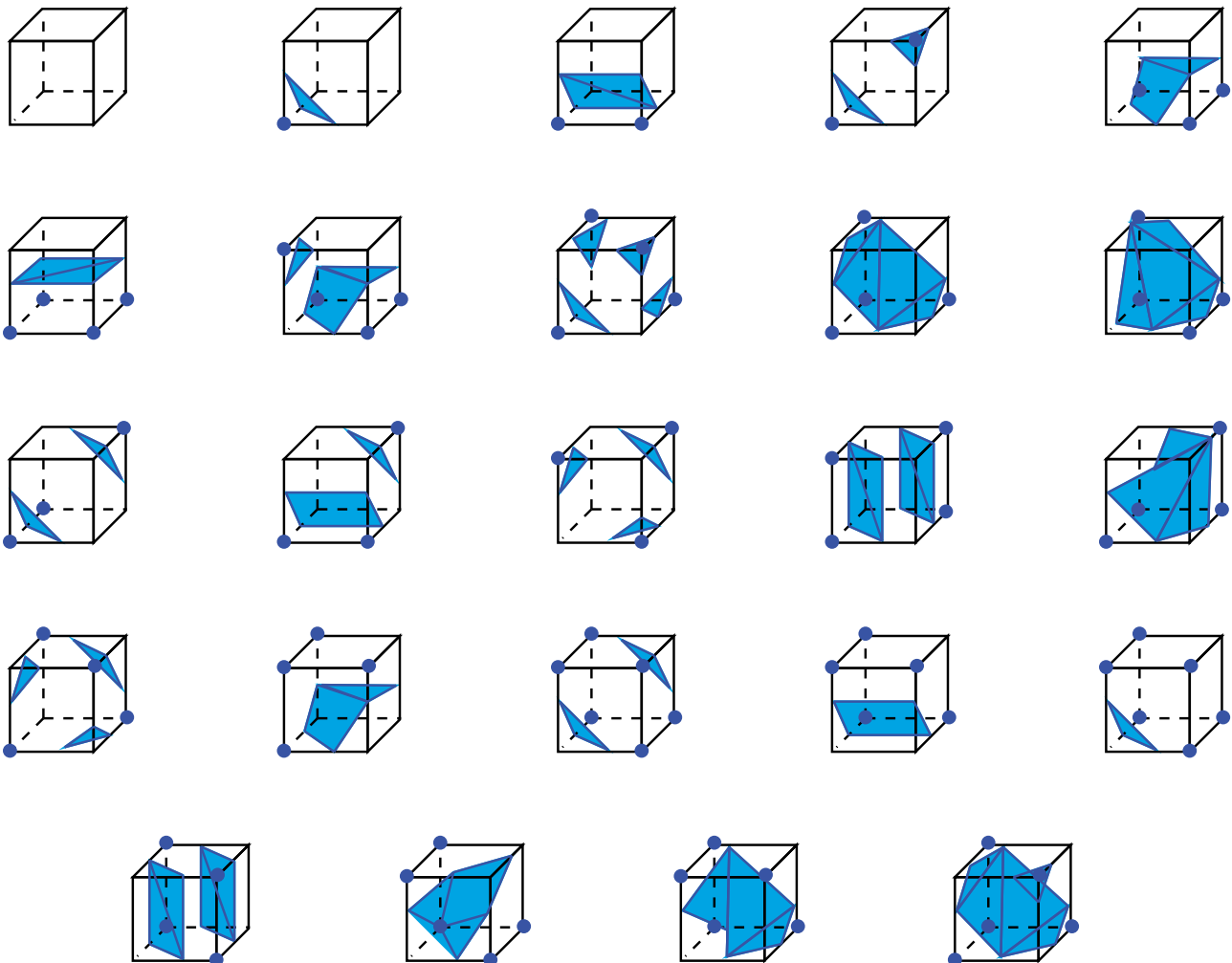


Figure 4. The expanded topology configuration of marching cubes algorithm (M3C).

images with corrected transformation. During the image preparation step, some different options could be used. In this step, the small region can be removed. The small grains removal option

consists of removing small disconnected or diagonally connected islands (smaller than a given number of voxels) to prevent small artificial isolated surfaces. This removal cannot fix critical

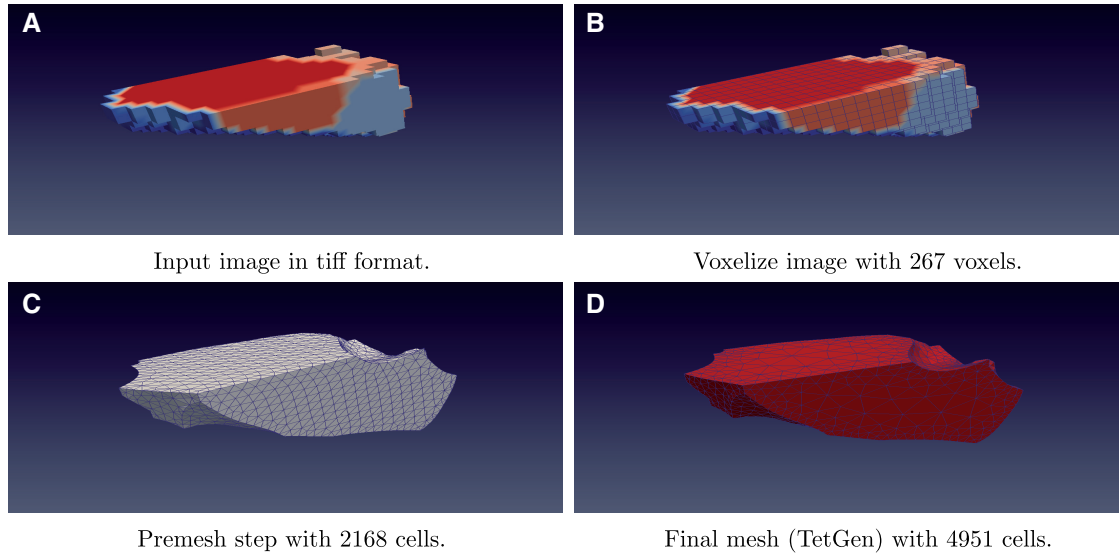


Figure 5. The volumetric tetrahedral mesher process on a given grain.

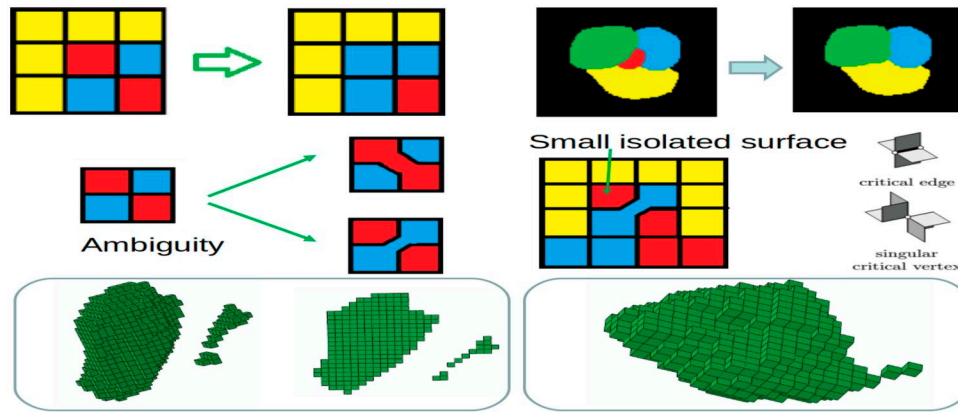


Figure 6. Small grains removal and critical voxel connectivity problems.

connectivity in cases of grain with a hole. Additionally, the connected voxels could be fixed diagonally. This means that elements up to a squared distance of connectivity from the center are considered neighbors. In some cases, it means that voxels with a common edge or vertex are in the same grain. Figure 6 illustrates semantic of the small grain removal and critical voxel connectivity problems.

A tetrahedral element possesses six dihedral angles, where each angle is associated with an edge that constitutes it. It is defined as the angle between two faces incident on it. Assume that each tetrahedron $abcd$ has vertices a , b , c , and d , with the following lengths, $u = b-a$, $v = c-a$, and $w = d-a$. It is easy to see that the dihedral angle θ at edge ab is the angle between these two normals, given by the inverse cotangent formula,

$$\theta = \cot^{-1} \left(\frac{(u \cdot u)(v \cdot w) - (u \cdot v) \cdot (u \cdot w)}{\sqrt{u \cdot u} [u, v, w]} \right), \quad (2)$$

in which $[u, v, w]$ means the scalar triple product operator of vectors u , v , and w . It could be minimized or maximized to determine its extrema. In the following, let the total set of points be partitioned into two groups. The first is the fixed points P_{fix} , and the second is the movable points P_{mov} . Recall that the original

version of the Laplacian algorithm consists of replacing the position p_i of point i with the average of the locations of adjacent points, as shown in Fig. 7. Formally, we have the following for every new position:

$$p_i = \begin{cases} q_i, & i \in P_{\text{fix}} \\ \frac{1}{|adj(i)|} \sum_{j \in adj(i)} q_j, & i \in P_{\text{mov}} \end{cases} \quad (3)$$

There are two approaches or techniques to compute new positions p_i of any vertex i . The first situation is to change all positions q by p in only one step. This technique supposes that the vertices obtained in each step from algorithm iteration are the same. Then, the new position p_i is entirely the set of positions, namely q . $Adj(i)$ denotes the set of adjacent vertices of one vertex i . Normally, the new position p_i depends on the positions of its adjacent vertices $j \in Adj(i)$. The convention of fixed point states that if $Adj(i) = \emptyset$, then $i \in P_{\text{fix}}$. After iteration of the algorithm, the previous points q_i go towards the new positions p_i , which are produced by the Laplacian algorithm. The second situation occurs when the new points are modified compared to current points q_i . We say that vertices were moveable while the topology of the mesh remains unchanged. Then, the vertex's location in the FE

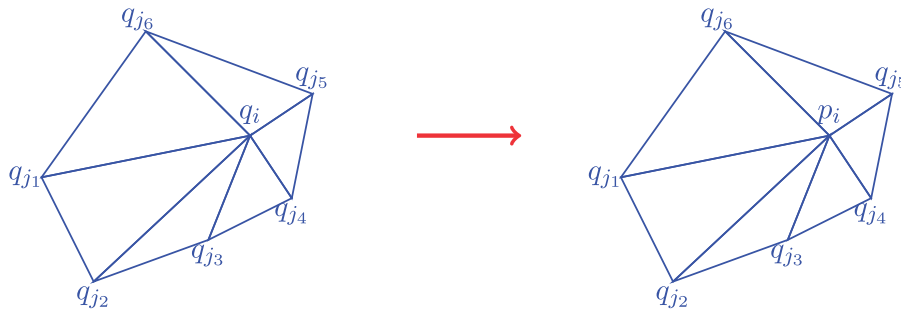


Figure 7. The Laplacian algorithm.

mesh is corrected by computing a new position as the average of the locations of vertices in the neighborhood on the mesh.

Here, the basic idea of the smoothing method predefined previously as the third stage is justified as follows. We have applied the smoothing technique based on the Laplacian smoothing approach. This approach, described in [74, 75] is defined as smoothing unstructured vertex sets. It is considered a new way to reconstruct a new surface mesh with high performance. The effect is to correct the location of vertices searching for further topological correction. Insisting here that a considerable effort is made to control minimum dihedral angles during this procedure for the non-manifold surface. Liang *et al.* [76] described the advantages of applying this technique to dihedral angles to ameliorate the degree of smoothness in quality.

Note that B-splines are a class of polynomial pieces with a particular connection between vertices that are used to create smooth curves. These functions could be first degree, second degree, or higher, and overlap with each neighbor. In the current work, weighted B-splines are used to guarantee the smoothness, particularly concerning multi-materials edges. In the same context, related works [77–79] are briefly reviewed. It is comfortable to use B-splines from an existing Python library named `scipy.interpolate`. The `splrep` function allows specific determination of the B-spline representation to define the control coefficients. It is aimed at developing representation for spline curves by constructing curves that interpolate these points. The first example is subjected to all the microstructure, in which B-rep of degree 3 is used to represent it geometrically. Figure 8 illustrates the use of an accurate B-spline based on the initial surface with the proposed modified criteria of our algorithm. This explains the displacement of the initial edges over resulting curves. The final diagram suggests the superposition of both these surfaces. This smoothing problem is solved by using three iterations to improve weights, prevent self-intersections, and compare their trajectory.

Meshing robustness in some material types

In this section, we investigate the reactions of some parameters involved in the M3C algorithm on the mesh information. Two materials are studied: composite and polycrystalline. This investigation is classified into two groups. The first one occurs while generating the initial surface mesh. The second one takes place during remeshing process, and it is aimed at knowing how the resulting surface is sufficiently smoothed by modifying each parameter. The mesh size panel characterizing the mesh quality is described by specifying information concerning cells. Then, the number of faces, edges, and corner nodes is reported in each studied case.

The first test consists of removing only small regions, especially those with 100 voxels or fewer forming the input image. It is presented in Table 1, performing the generated mesh by 0.2% for both material types in quality. This option is advocated for affecting accuracy in mesh size.

The next test examines fixing diagonal connectivities of voxels while reading the input image. This test is illustrated in Table 2, demonstrating that this parameter does not influence mesh refinement, consequently providing the same quality mesh.

In what follows, the objective is to successively remove small regions from a given 3D image. This is done by progressively removing 64, 128, and 1000 voxels in each case, as presented in Table 3. Removing the same voxel number affects the polycrystalline material much more than the composite material. It is justifiable to delete some voxels by utilizing this option for modelling polycrystalline microstructure with high accuracy during the image preparation step.

The contribution concerning the connectivity paradox for solving the problem of edges and surfaces is presented. This solution is considered novel for conserving the morphological properties of the structure when it is applied. Connectivity controls cells that are part of the structure. It ranges from 1 to 3, which is the default value. This value means that voxels with a shared edge or vertex are in the same phase. It allows for the improvement of mesh quality in the smoothing process. This study is detailed in Table 4.

It is feasible to bring correction to the surface topology if critical connectivity is not fixed or not regular. If this option is activated with a true Boolean value, the obtained mesh gets the same surface and edge nodes for both material microstructures, although, the corner nodes differ in small numbers. This variation is explained by the fact that new points belonging to the cell center are appended to some facets. The resulting test is summarized in Table 5.

The smoothing operation is performed during the remeshing steps. If the iteration number is assumed to be large, then the generated mesh enlarges significantly, as shown in Fig. 9. The following operation is the implicit fairing approach [80]. This approach is based on Laplace smoothing, which can be regulated and ranges from 0 to 1. It is adopted for naturally controlling the volume of a small component in the region of the element. If this parameter takes a value near from 0 or 1, the mesh quality is improved for composite material. However, the meshing process of polycrystalline material appears to be dissimilar. This evoked test is illustrated in Table 6 and is clarified as follows. When increasing this coefficient, the number of nodes and cells declines, consequently degrading mesh element quality.

Some parameters, such as distance and angle, affect mesh quality. The first one is utilized for the moving distance

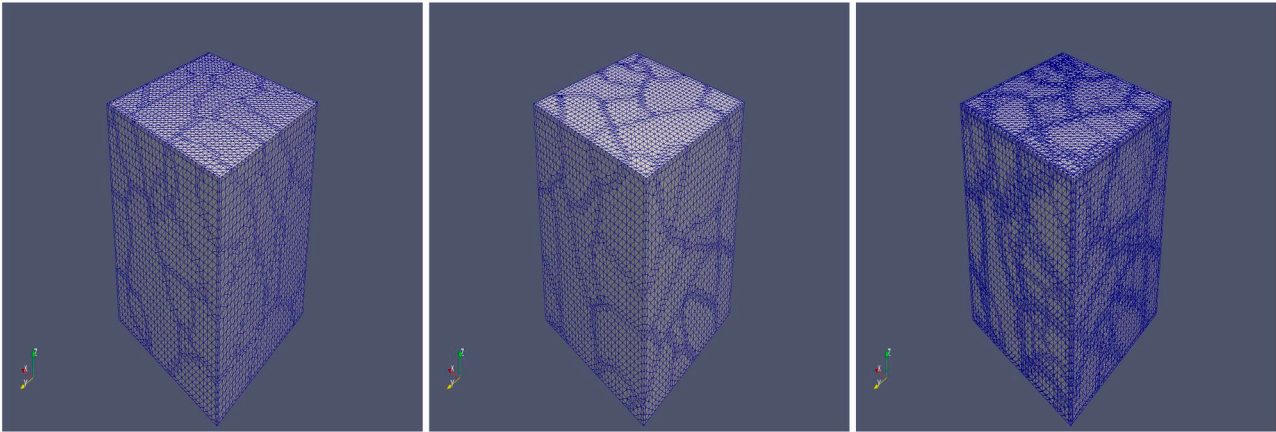


Figure 8. Smoothing weighted B-spline approximation for one given microstructure.

Table 1. Mesh characteristics in removing small voxels

Material types	Option	Surface nodes	Edge nodes	Corner nodes
Composite	False	501 455	14 385	154
	True	500 259	13 270	96
Polycrystalline	False	17 886	4706	545
	True	17 042	3953	351

Table 2. Mesh characteristics in fixing diagonal connectivities of voxels

Material types	Option	Surface nodes	Edge nodes	Corner nodes
Composite	False	500 256	13 274	97
	True	500 259	13 270	96
Polycrystalline	False	17 040	3953	351
	True	17 042	3953	351

Table 3. Mesh characteristics in removing voxels

Material types	Option	Surface nodes	Edge nodes	Corner nodes
Composite	64	500 259	13 270	96
	128	499 115	12 643	78
	1000	497 699	12 263	67
Polycrystalline	64	17 042	3953	351
	128	16 123	3404	269
	1000	10 569	1267	46

Table 4. Mesh characteristics in voxel connectivity

Material types	Value	Surface nodes	Edge nodes	Corner nodes
Composite	1	499 541	12 793	81
	2	500 229	13 236	94
	3	500 259	13 270	96
Polycrystalline	1	17 036	3949	348
	2	17 037	3949	348
	3	17 042	3953	351

Table 5. Mesh characteristics in correcting the surface topology

Material types	Option	Surface nodes	Edge nodes	Corner nodes
Composite	False	500 259	13 270	99
	True	500 259	13 270	96
Polycrystalline	False	17 042	3953	366
	True	17 042	3953	351

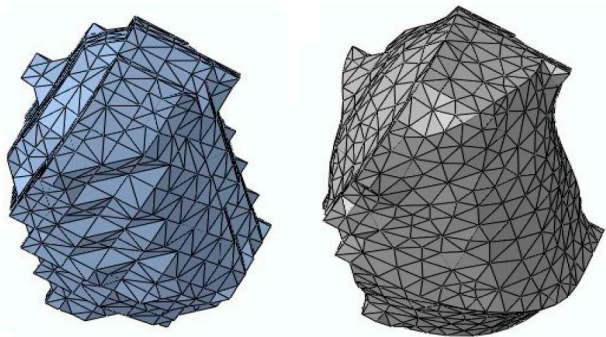


Figure 9. The left diagram is with smooth iteration equals 2, and the right diagram is with smooth iteration equals 200 for selected grain from polycrystalline material.

Table 6. Mesh characteristics in correcting the surface topology

Material types	Value	Total nodes	Total cells
Composite	0.1	113 062	237 157
	0.5	111 297	233 627
	0.9	113 064	237 161
Polycrystalline	0.1	17 886	39 995
	0.5	17 543	39 309
	0.9	17 133	38 489

limitation, which refers to the maximum allowed distance to move the vertex during the smoothing (1 is the size of a voxel). Running the test consists of jumping this variable from 0.2 to 2. Then, the quality mesh remains practically the same (see Table 7) for both materials. The second one is designed to mean a minimal dihedral angle in order to control flip and collapse operations. It is aimed to generate any degenerate triangles. Jumping this parameter from 10 to 20 involves minimal repercussions in the generated grid mesh for both materials, as presented by Table 8. The third one is used for the distance limitation. It is defined as the highest distance admitted between two mesh points during the smoothing process. Table 9 illustrates that when varying this parameter from 0.01 to 0.1, the number of cells decreases. However, it knows an evolution while it jumps from 0.1 to 0.3 for composite material. Reducing it will of course cause the growth of cells for polycrystalline material.

These invoked algorithms, which are involved in mesh generation workflow, absorb much time. Here, we investigated fully the numerical study of optimization procedures in some

Table 7. Mesh characteristics in moving distance limitation

Material types	Value	Total nodes	Total cells
Composite	0.2	111 297	233 627
	2	111 324	233 681
Polycrystalline	0.2	17 543	39 309
	2	17 540	39 303

Table 8. Mesh characteristics in minimum dihedral angle

Material types	Value	Total nodes	Total cells
Composite	10	111 243	233 519
	20	111 297	233 627
Polycrystalline	10	17 545	39 313
	20	17 543	39 309

Table 9. Mesh characteristics in minimum shifting distance limitation

Material types	Value	Total nodes	Total cells
Composite	0.01	112 294	235 621
	0.1	112 155	235 343
	0.3	111 297	233 627
Polycrystalline	0.01	18 337	40 897
	0.1	17 560	39 343
	0.3	17 543	39 309

variables involved in the mesher. It was demonstrated how these computation efforts impact simulation and workflow performance. For instance, take the polycrystalline microstructure, assuming that the input image is constituted with numerous grains. Next, removing all of them facilitates notable calculations. During this step, it is shown that no impact on mechanical behavior is anticipated. In addition, image quality and resolution stay adequate. One more dominant step of the image preparation consists of eliminating voxels, diagonal connectivity. Another possible way, considered more accurate, is the smoothing method. The initial surface is smoothed and a half-edge data structure is created. In these methods the interface mesh is refined, centered on the modified Laplacian smoothing approach and keeping a minimal limit of dihedral angles. The principal improvement during the smoothing procedure is to also use weighted B-spline as a smooth approximation of multi-material edges.

Spherical inclusion

This section is devoted to determining how the M3C algorithm reacts with geometrical object curvature, and if it exists, its influence on microstructure geometry. In particular, the spherical inclusion is generated into a cubic box matrix, setting a volume fraction target and iterating until the expected value is reached.

In the following, two main examples are studied. The first consists of adding only one sphere inside the cubic box matrix, as shown in Figs 10 and 11. The second involves generating random spherical distribution of inclusion with a radius of 0.08. Here, there are two phase numbers, in which zero designs the matrix, and one designs the inclusion. For this, L_{max} is introduced as the parameter that controls the upper limit of edge size. This variable is important for the volumetric pattern found in the image (inclusion, pores, defects, etc.), and needs to be sufficiently small to capture the size of these patterns. It is involved in the smoothing

and remeshing stages. The aim here is to understand how it acts on the curvature of some objects during these steps.

For one spherical inclusion given in Fig. 10, the quality mesh between the voxel image and its mesh generation is interpreted. For this, the volume fraction between them is compared. The resolution is set to 100 voxels in each dimension. Additionally, the volume fraction reaches nearly 0.4% in this case. The mesh generated by the M3C algorithm depends on the initial image resolution. We can see in the resulting mesh that the volume fraction is equal to 0.38%. There is not much information lost in the mesh. Note that L_{max} does not react in the same way when there are one or many objects inside the box. For one distribution in Fig. 11, the quality mesh is better when the value of L_{max} rises, even in some places that do not present the real format. However, the quality is better, as shown in Fig. 12, when the value decreases considerably.

The main challenge is to investigate the homogenization of elastic properties with periodic boundary conditions (PBCs). The reader is referred to the latest works [81–83] in which mean-field homogenization techniques are described. A direct homogenization approach consists of utilizing the discretization of finite element boundary for a given cell supposed to be a unit. The effective stiffness matrix of any microstructure is determined with computation, and as a mathematical association between macro stress and macro strain. The objective here is to determine values for the effective stiffness tensor macroscopically anisotropic, concerning a two-phase system containing spherical inclusions, as illustrated previously in Fig. 11. Notably, the authors in [84] improved the homogenization method to obtain the macroscopic behavior of a composite material containing inclusions. The method of computation is given as follows. First, the meso-level is solved numerically, obtaining consequently effective properties of the reference cell. The mechanical equivalence between both levels includes using the averaged volume of strain. These solutions are used as data for the macro level. In the frame of this study, the example below illustrates the computation of the effective stiffness tensor (GPa) in the case of tetrahedral mesh, with different values of L_{max} . Mainly, the extraction of effective stiffness from a post-processing of FE analysis is made. Note that in comparison, some values of the matrix deviate, but some of them are close to each other.

$$\begin{pmatrix} 3.19192322e+03 & 1.34579817e+03 & 1.35212830e+03 & -2.64233174e+00 & 7.90282378e+00 & 1.44751926e+00 \\ 1.34621761e+03 & 3.25753027e+03 & 1.37153188e+03 & -3.10692435e+00 & 6.19223731e+00 & 7.89522376e+00 \\ 1.35224413e+03 & 1.37188695e+03 & 3.27879396e+03 & -1.82769612e+00 & 1.22187978e+01 & -2.20415569e+00 \\ -5.43503268e-01 & -9.69332255e-01 & 1.05735578e-01 & 9.40648369e+02 & 2.55148625e+01 & 1.58844643e+00 \\ 5.39540112e+00 & -1.53467618e-01 & 2.57406879e+00 & -2.65608678e+00 & 9.42483289e+02 & -7.00515308e-01 \\ 1.22312682e-01 & -3.89214626e+00 & 6.23974484e+00 & 3.42909799e+00 & -7.73431986e-01 & 9.94047956e+02 \end{pmatrix}$$

$L_{max} = 40.$

$$\begin{pmatrix} 3.34441025e+03 & 1.43375210e+03 & 1.43378574e+03 & 1.11131296e-02 & -2.74269583e-02 & 1.00857499e-02 \\ 1.43375284e+03 & 3.34434634e+03 & 1.43377921e+03 & 1.08251414e-03 & -2.93452658e-03 & 5.20068337e-02 \\ 1.43378624e+03 & 1.43377950e+03 & 3.34453395e+03 & 1.29711341e-03 & -5.58556418e-03 & 4.04267486e-02 \\ -1.53508958e-03 & -1.38225908e-02 & -5.80169809e-04 & 9.55292506e-03 & 1.02149408e-02 & -8.06979802e-03 \\ -1.90693377e-02 & 2.96526666e-03 & -3.10114180e-03 & 7.60937498e-03 & 9.55332224e-03 & 1.05433061e-03 \\ 8.10469053e-04 & 3.79066706e-02 & 3.64637700e-02 & -6.46185468e-03 & 1.73597824e-03 & 9.55315355e-02 \end{pmatrix}$$

$L_{max} = 2.$

Most of the computation time used to generate the resulting mesh is spent on the remeshing process. The aim of this work was to improve the implementation of all developed algorithms in the Python interface, including multi-material processing. This performance improvement is proposed in the M3C algorithm, which is illustrated as follows. The discrete step of looping over all the mesh elements was performed. This looping reduces operations such as edge splits, edge collapses, and edge flips, which require significant time to implement. Furthermore, the reduction of operations fixes a manifold interface that connects

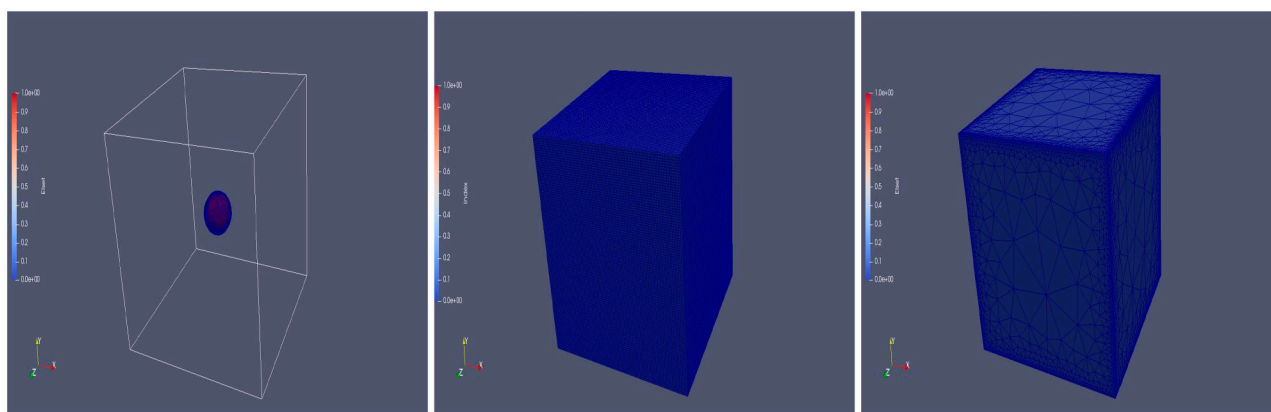


Figure 10. Simple case for one spherical inclusion. The left diagram presents the schematics. The middle diagram is image preparation containing 10^6 voxels, with 0.4% volumic fraction. The right diagram presents the tetrahedral mesh generation, with 0.38% volumic fraction.

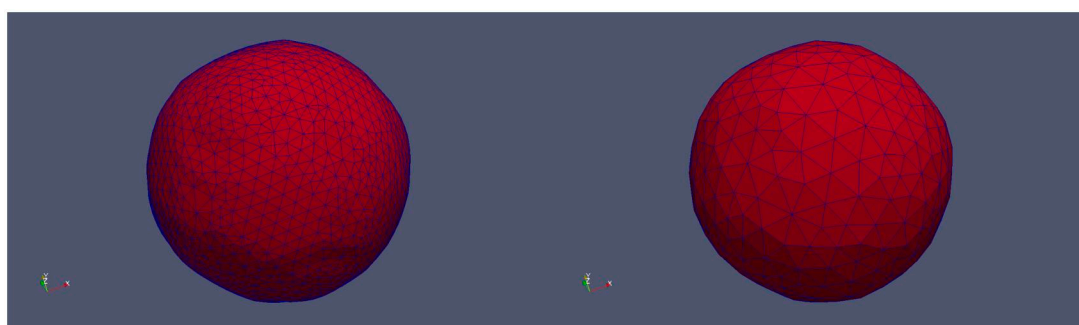


Figure 11. The left diagram shows that volume fraction is 0.38% when L_{max} equals 40, and the right diagram shows that volume fraction is 0.37% when L_{max} equals 2.

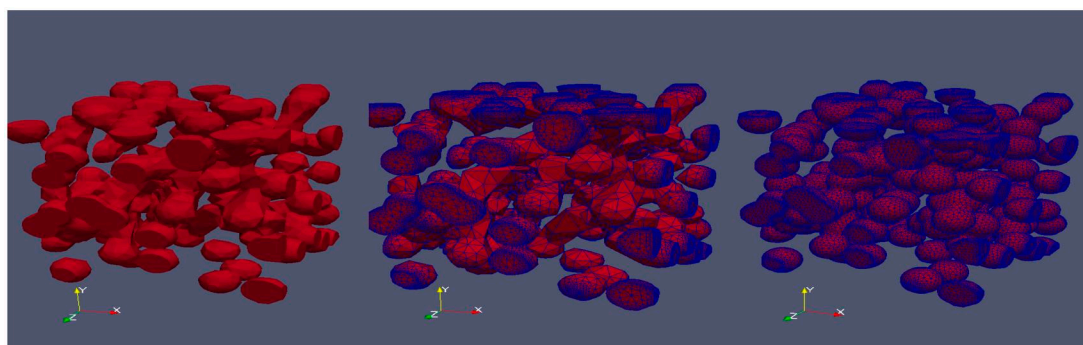


Figure 12. The left diagram is the input image containing 10^6 voxels, with 30.02% volumic fraction. The middle diagram shows that volume fraction is 23.04% when L_{max} equals 40. The right diagram shows that volume fraction is 28.62% when L_{max} equals 2.

components of the material. For instance, if a bi-material is considered, then a set of connected edges and corners that are shared between the material regions is repaired. The aim of this repair was to realize the multi-material segment processing for decreasing the vertex count. Then, the dihedral angle will be limited, and the corresponding nodes from the edges will be controlled to guarantee good-quality elements for FE analysis. This quality is also improved during the smoothing process by reducing the node count of the smoothed mesh. The produced corners are usually fixed during the smoothing process. In addition, multi-material segments are smoothed separately from the surface nodes, and the surface is smoothed using selected multi-material segment nodes. Some mesh characteristics are controlled during the smoothing process, such as the minimal dihedral angle between the minimum and maximum value.

The originality of the current work lies in how the multi-material marching cubes algorithm is improved and utilized for the initial mesh reconstruction. One of the typical improvements is that the Laplacian smoothing algorithm is adjusted to process the non-manifold reconstruction mesh. This new reconstruction averts surface self-intersection and controls dihedral angles during this step of thin and sharp regions. The smoothing operation and flipping edges are illustrated in Fig. 13 by varying the control parameters such as smoothing speed and angle. The details of the newest implementation provided in the M3C algorithm are described as follows. The main improvement of this implementation focuses on the LUT-based performance. This implementation distinguishes four cases in which the construction of a LUT is considered. The first case consists of cells with exactly two materials. Then, the LUT has around two hundred fifty possibilities, as in the classical MC

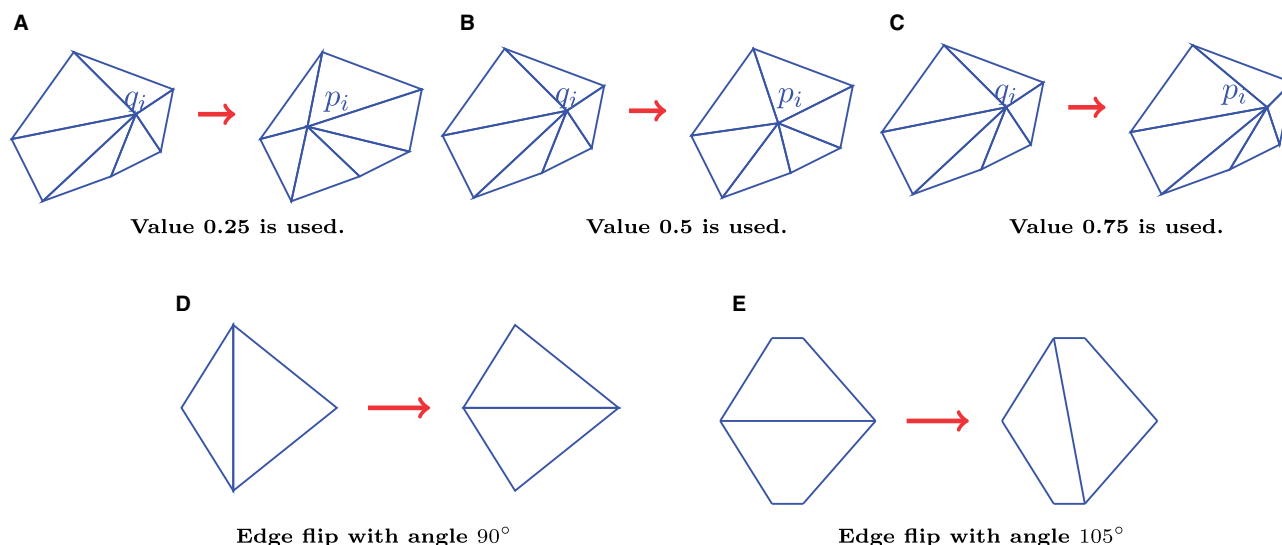


Figure 13. The Laplacian algorithm with local mesh operation. The smoothing aspect is presented with different control parameters 0.25, 0.5, 0.75, which are shown respectively in diagrams (A), (B) and (C). Edges flips with different angles are shown in diagrams (D) and (E).

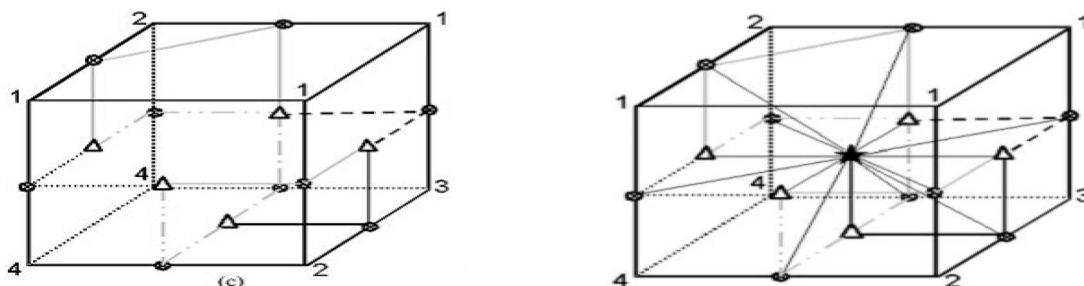


Figure 14. The nodal singularity showing the interfaces of bi-material connected only in one point.

algorithm. This situation is the most difficult algorithmically due to the RAM limitation. Cells with more than two materials but without centered facets are considered in the second case, where there are eight triangles per cell. Cells with exactly two centered facets are considered for the next case. Finally, cells with more than two centered facets are taken into account for the final case. This operation consists of the cell mesh being viewed as a merging of a six-facet mesh. A facet mesh is obtained from the direct LUT with sixty possibilities where a maximum of four triangles may occur.

Problems arising from the M3C algorithm due to diagonal voxel connectivity may exist. Globally, it is absurd to automatically guarantee the diagonal connectivities for any given 3D imaging data, but the developed version of the M3C algorithm works correctly in many cases. Our main work is to correct the topology due to this diagonal connectivity if one common voxel vertex occurs. Additionally, the edge intersection of the resulting surface in the case of two diagonal connectivities in a single cell is fixed by removing this connectivity. The original M3C could be slow, because each cell is meshed individually (LUT algorithm is not used). Our implementation is based on the LUT approach as described previously in the case of bi-materials. The multiple material marching cubes ambiguity is solved when the interfaces of bi-materials are connected. This connection occurs only at one point, consisting of the nodal singularity as shown in Fig. 14. The surface stays free from edge singularity.

The divide-and-conquer is an important algorithm strategy that has been developed for the mesh. This algorithm is used in

the image processing and one part of the M3C algorithm. If the divide-and-conquer algorithm cannot be used, then the cell meshing. The aim is to guarantee the conformity of the partition pattern on each face taking into account the neighboring cell. There are three cases for meshing this cell: cells without centered facets, the cells with exactly two centered facets, and cells with more than two centered facets. In the third situation, a cube-centered node is generated and reserved for all the material identification of the cube. These three possibilities are summarized in Table 10.

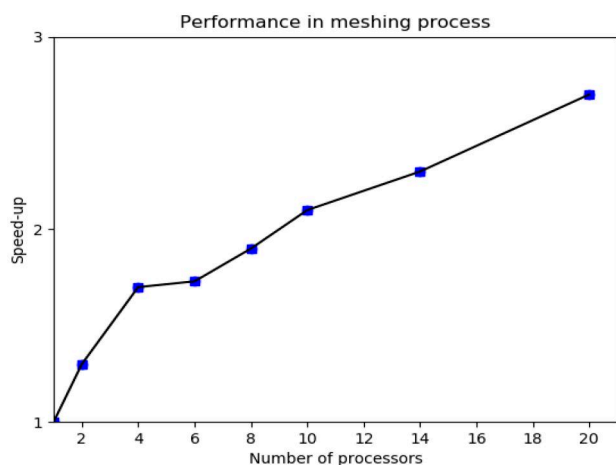
The isosurface method with crack patching is adapted in this work when the intersection occurs between two cells of different materials. The count of triangles is reduced when this method is applied to the marching cubes algorithm. This algorithm becomes ideal in terms of topology consistency since this topology is preserved as described in many works [85, 86]. Our contribution is to solve face ambiguity by determining whether sharp features overlap. Then, all different materials are bounded by triangular patches. We will demonstrate in the next section the effectiveness of our algorithm on several possible applications using different image resolution models.

Computation chain on Abaqus

This section is dedicated to developing a framework based on tetrahedral meshing with material properties in the .inp file that is supported classically by Abaqus. For this purpose, a new script is

Table 10. Cell meshing while no using the divide-and-conquer algorithm

Mesh information	No. centered facets	Two centered facets	More than two facets
Max no. surfaces	4	5	24
Max no. triangles in a surface	5	6	1
Max no. triangles in a cell	6(8)	12	24

**Figure 15.** Speed-up of time meshing in terms of number of processors.

implemented. This routine takes as an input a set of interface meshes (one or several disconnected grains for each material orientation) with the following generation set of tetrahedral meshes, merging them in single model and storing the resulting mesh in this format.

An efficient finite element method implementation needs the domain decomposition necessary for parallel computing. This implementation consists of partitioning the domain into subdomains. Meshing with partitioning usually offers many advantages and is expressed by distributing the domains over all the n MPI processes. The elementary stiffness matrix is assembled into one global linear system (large and sparse matrix), which is solved practically with iterative techniques in parallel. Our framework is not based on this invoked approach since the chain process is derived exclusively from Abaqus solver. Nevertheless, the process is parallelized for meshing operations by dividing interfaces along boundary edges. Additionally, the model is attended with all required input .inp files, and executed by submitting the job in the parallel cluster. Figure 15 depicts the parallel implementation and performance testing of a given unstructured grid.

The grid mesh was generated previously using the M3C algorithm and written in this format, illustrated by the element sets for different parts of the geometry. The microstructure studied concerns composite and polycrystalline materials, hemp with basalt, and glass fibers. The aim here is to show the ability of this framework to be functional under Abaqus.

After obtaining the grid mesh with corresponding element sets, it is interesting to compute the volume of each element. The computational method for the volume fraction is implemented to calculate a mixture's composition for each given microstructure. The volume of a tetrahedron defined by its four points is calculated to generate the volume fraction of each element set by grouping the values of the phase.

The first material microstructure investigated is the polycrystalline one which is shown in Fig. 16A in tiff format. It is presented as a set of $25 \times 25 \times 25$ voxels containing 87 grains labeled from 1 to 87. Next, the input file is loaded, and the necessary steps are followed to run the mesh solver. Simultaneously, we are making rotations/flips for correct positioning in the simulation system axis. The generated mesh is illustrated in Fig. 16B as a vtk unstructured grid. The same tetrahedral mesh in Abaqus inp format is shown in Fig. 16C.

The mesh generation relies on an optimization algorithm fixed in analogy between a good-quality finite element tetrahedral mesh, and the initial voxel geometry. The methodology proposed here is to exploit the advantages of using conforming finite element mesh and correlate it with the voxel-based discretization one. This method focuses on local volume fraction control. Figure 16D illustrates a quantitative comparison of the computed volume fraction between the voxel and conformal mesh results. For some grains, the tetrahedral volume fraction is greater than the voxelized one. These include the grains whose numbers 1, 3, 4, 8, 9, 11, 12, and 87. Some grains, such as 2, 5, 7, 10, and 13, have superior voxel volume fractions. Furthermore, the obtained tetrahedral mesh is considered to be of high quality, since there are not many analysis warnings, around 0.011%.

To inspect the obtained results of this simulated microstructure, the required output data are generated in .odb format. For this, the model is executed by submitting the job. The solver interprets how the job will be run based on a set of instructions contained in the given file. Since the model is supposed to be fully linear elastic, it follows that three specific sets of material properties are needed. The material properties are defined with the elastic constant of cubic crystal (Ni-Superalloys), given by the following values 219000, 157000, and 109000. Note that this crystal should be oriented through the microstructure. The linear problem is analyzed with steps in Abaqus.

Over this main simulation, even small polycrystalline aggregates are explicitly taken into account to bring a mechanical deformation response. There are generally six components of stress, denoted S_{11} , S_{22} , S_{33} , S_{12} , S_{23} , S_{13} . Figure 17 depicts the results for anisotropic polycrystalline material. It exhibits a real state of stress that corroborates well with the analytical solution from the theory of elasticity to validate this model. These stress components S_{11} and S_{22} on representative volume element (RVE) are around 10^5 MPa. Additionally, the maximum Von Mises stress provides 3.610^5 MPa, to forecast the yielding of this material once subjected to complex loading. This microstructure has a high anisotropy that underlines its excellent properties to gain in orientations.

Since the investigated material is the composite material, the influence of elastic anisotropic material is still illustrated. In what follows, this study examines the microstructure and mechanical properties of the selected segmented with 3D images presented in Figs 18 and 19. The aim is to mesh this microstructure from its individual phases (pores, yarns, matrix). Although generating tetrahedral mesh works well for polycrystalline material, its robust generation is unknown for a composite material, as shown in Section "Meshing robustness in some material types". The image shown in Fig. 18A, contains 8.10^8 voxels, and its microstructure consists of randomly distributed fibers. It is distinguished by five phases, labeled from 0 to 4. Next, it is meshed in vtk format, viewed under the ParaView tool, as shown in Fig. 18B.

The resulting volumetric mesh in Abaqus inp format with element set definition for labels is presented in Fig. 18C.

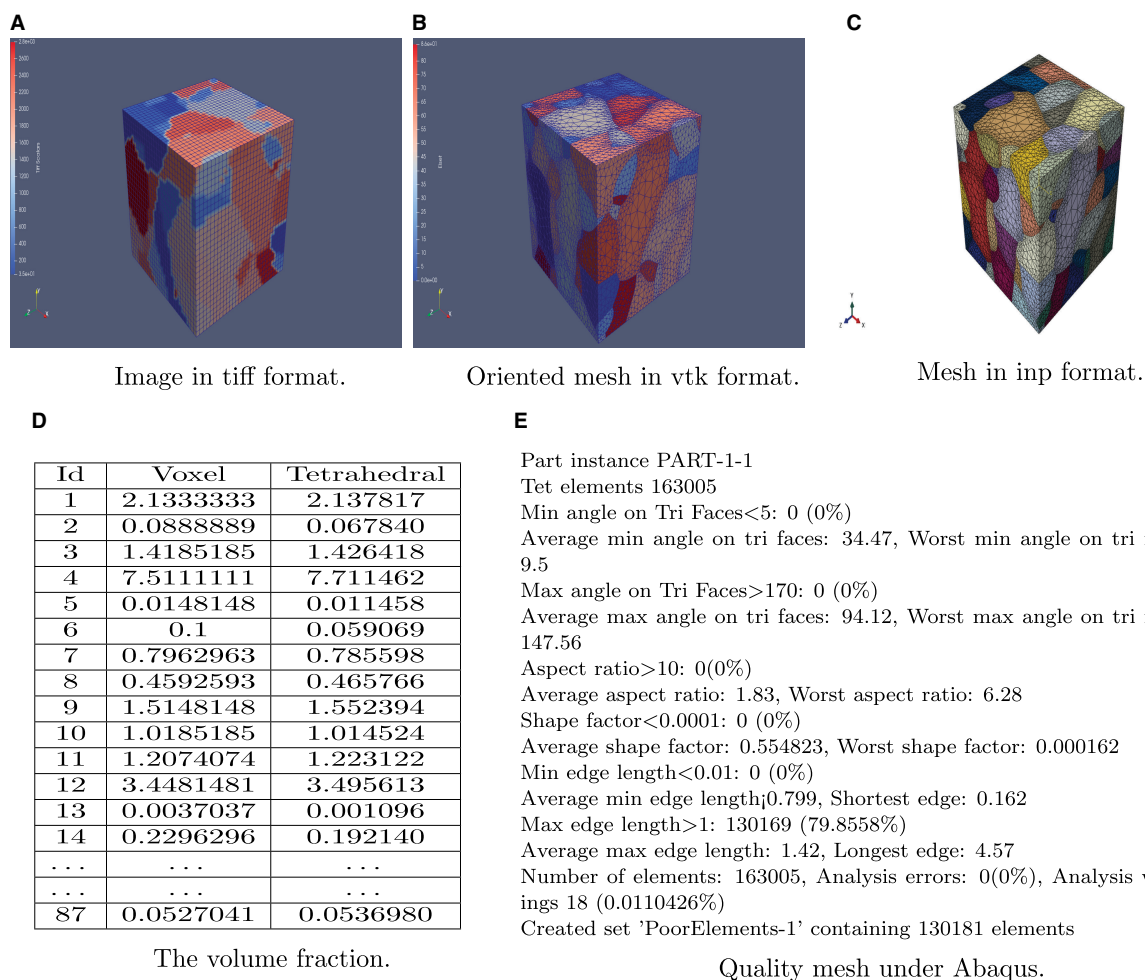


Figure 16. Generating conformal mesh and quality mesh for polycrystalline material microstructure with multiple material marching cubes algorithm (M3C).

Additionally, Fig. 18D presents results from a numerical study of the influence of fiber volume fraction when composites with 3D texture are covered. It illustrates that tetrahedral elements dominate in the majority of the volume fraction.

The development is continued as part of image-based modeling for 3D woven composites with very thin matrix areas. The periodic boundary conditions (PBCs) have been chosen during the modeling process in this particular study. Indeed, they are properly defined in Abaqus viewed as a robust package used for simulation of composite materials. The aim is to perform the simulation from segmented 3D images in Abaqus with these conditions. The FEM mesh given in Fig. 18 is used to interpret strain distributions of this microstructure. This example of the variation of the Green-Lagrange strain E_{11} is shown in Fig. 19, following the direction aligned with the x-axis. Next, the same strain E_{11} in all steps. During these loading steps, there is an axial extension in the second step following the y-direction and the third step following the z-direction. Note that shearing concerning the remaining simulations. This strain E_{11} amplitude usually takes the weak value, and can be positive or negative. The deformed shape and contour plot of this strain is consistent. Indeed, these conditions under Abaqus are enforced at node sets located on the surface, which are used for applying concentrated loads. The nodal boundary forces are created to show the capability of capturing geometric variations. Finally, the required displacements results correspond well to the imposed ones across element

boundaries. During this numerical test, the elastic behavior could be viewed in very good approximation, while the composite structure deformation provides this natural response. This study is carried out to show that composite is more substantial than conventional material, specifically polycrystalline.

To make the best use of fiber-reinforced composites, it is crucial to have a good understanding and predictive ability of their micromechanical behaviors under ideal conditions. Natural fibers are generally suitable to reinforce plastics. In particular, natural fiber like hemp has excellent tensile mechanical and thermal properties. However, its potential is not based on fiber-reinforced composites. This may be explained by dislocations of presented fiber. The key idea about how modeling the constitutive the behavior of these composites is to use the Abaqus mechanics system, which has the potential to manage 3D fiber-reinforced composites. On the micro-scale, fibers and the corresponding fiber orientation in each phase are treated and taken into account. In this contribution, a framework for simulating anisotropic fiber composites has been considered explicitly in the numerical modeling. In particular, since fibers are randomly distributed with different orientations, then the related Euler angles have been used following the form (1).

Treatment of large images remains heavy for calculation and optimization depending on the complexity of composition. In this situation, the extraction of pavement becomes the best and most suitable option. The primary purpose of this approach is to

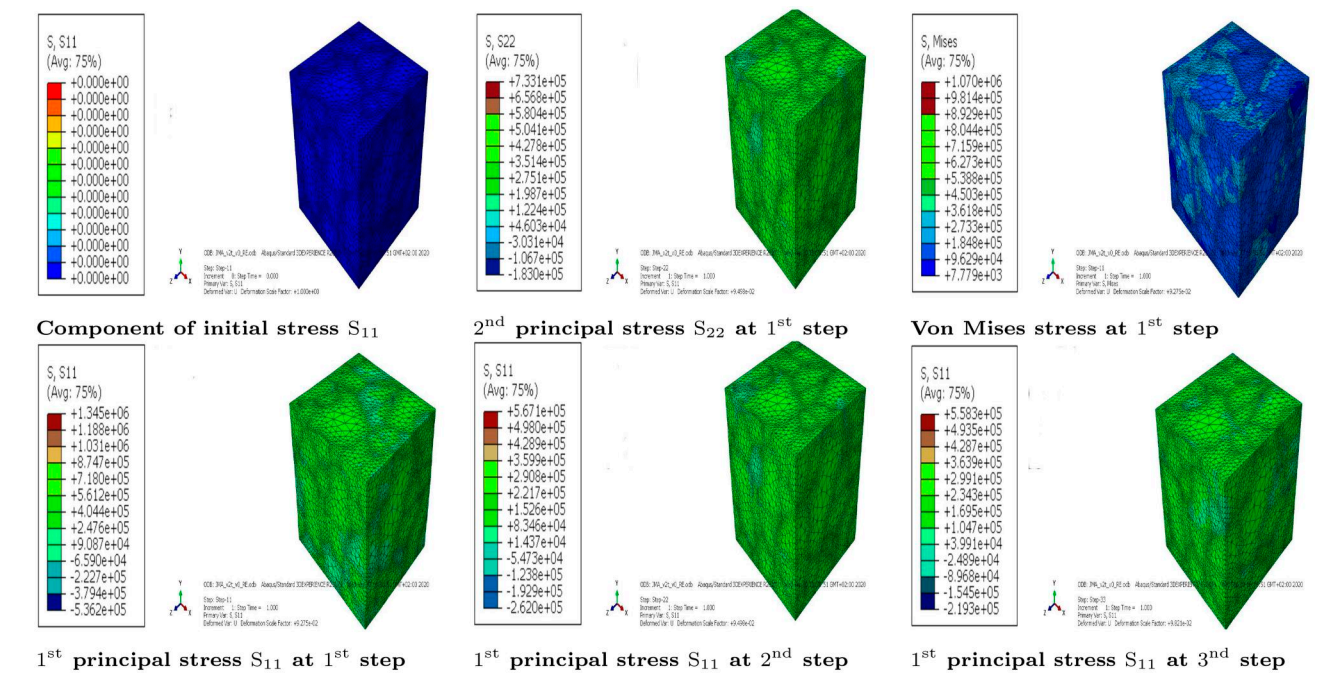


Figure 17. Running polycrystalline material under Abaqus.

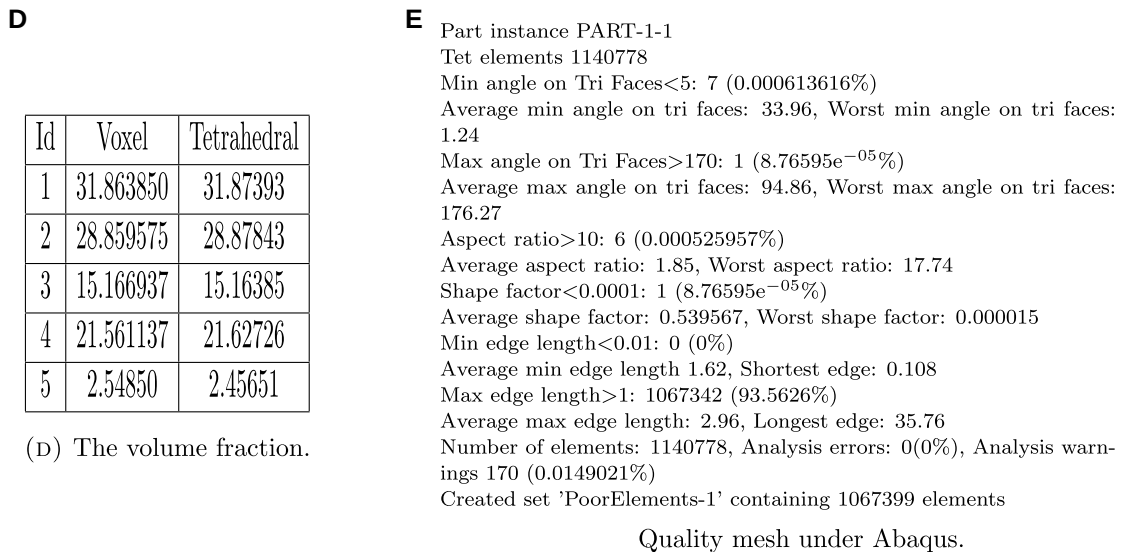
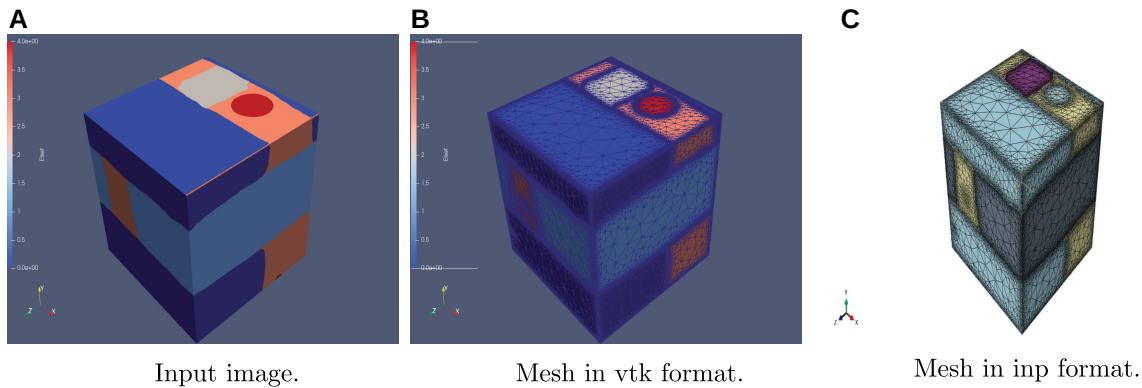


Figure 18. Generating conformal mesh and quality mesh for composite material microstructure with multiple material marching cubes algorithm (M3C).

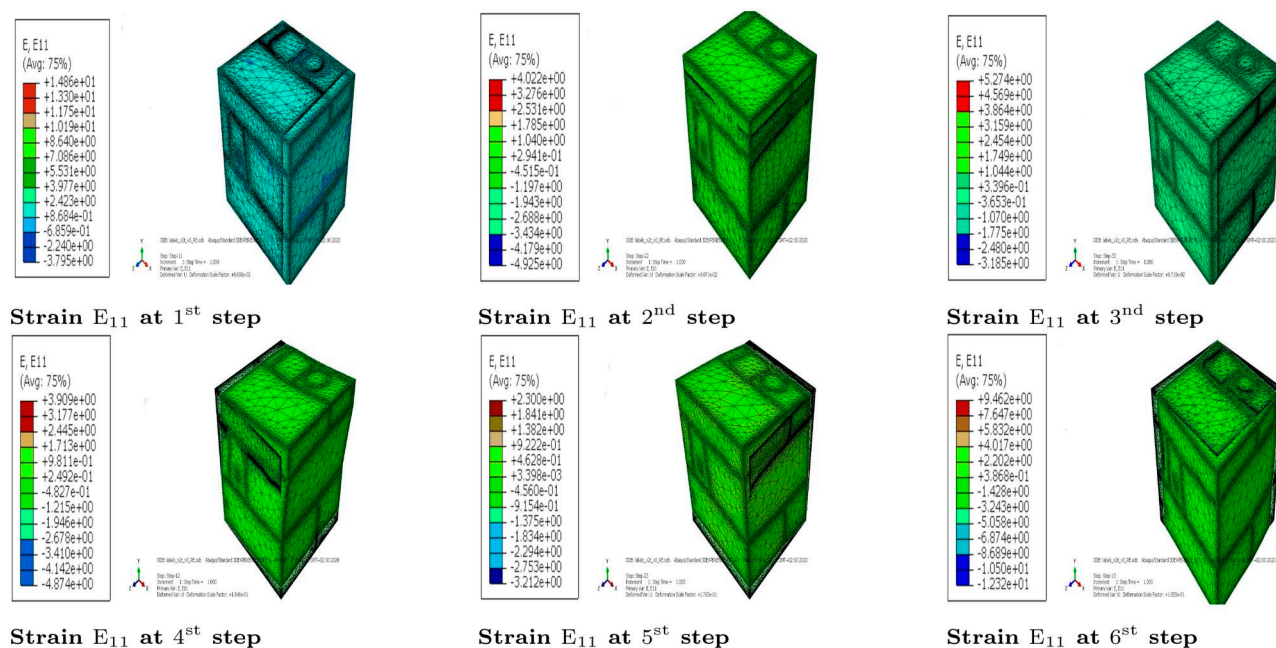


Figure 19. Running composite material under Abaqus.

optimize computations to reach greater robustness and faster execution. This approach is applied in this study, since the initial image is very heavy to be treated at around three billion voxels. A different type of composite is investigated in this work. It concerns hemp and basalt fiber reinforced composites. The original image is extracted into a small one, which is constituted with 8.10^3 voxels. As shown in Fig. 20, this microstructure is characterized by mechanical properties of unidirectional fibers, strengthened with resin composites loaded in the axial axis.

Detailing here the conformal mesh and its quality for this microstructure summarized in Fig. 21, and distributed into five diagrams. The tetrahedral mesh model from a multi-materials Fig. 21A is generated in vtk format that is presented in Fig. 21B. The same resulting mesh in Abaqus inp format is shown in Fig. 21C. This mesh keeps a rational material interface and tiny features like thin fibers. The hemp fibers are well presented by the tetrahedral elements since their constituents are about 3.85%. The empty phase is dominated by the voxels, which is not needed in this simulation. In this case, the voxel volume fraction reaches nearby 96.14%. This mesh maintains its high quality, having fewer analysis warnings as illustrated in Fig. 21E.

Now, we can proceed to make an interpretation of the mechanical analysis in order to examine the generated results. Here, meaningful results are used to perform the model. Then, we focus on finite element modeling using Abaqus to study the behavior of this microstructure. The objective is to predict the load-displacement behavior. The zone concerned is the fiber and matrix interface. The first component of stress S_{11} is analyzed, as shown by its variation in Fig. 22. It demonstrates that the stress varies axially with the x-coordinate in the first step. Similarly, the enlargement is remarked during the next step following the y-direction until the shearing plane, which occurs in XZ plane at the final step.

Normally, the material properties depend on the tissue type used for hemp fibers, and various other factors. The elementary hemp fiber is identified as a multi-layered tube. Rotation angles are considered to be the main factors involved in this dependence. These class fibers are made up of single cells considered

elementary for composing the structure. The elastic modulus is determined in terms of fiber diameter. In the test simulation, it was approximately 43 GPa, when there was a little rotation of the fiber, and 65 GPa when the rotation was forbidden. The simulation results could be expected to be more reactive when there are important rotations.

Experimental results

This section is devoted to the concrete consideration of the interface between experimental and computational mechanics. We carried out some experimental tests to validate the correctness of the FE framework. These tests concern 3D compressive and flexural testing. Note that 3D printed specimens under these tests generally have three loading directions: F_x , F_y , and F_z . Various size voxels were used to get the compressive and flexural strengths. The material properties used in this study reflect the nonlinear constitutive characteristics of these specimens. This study's main purpose was to consider the concrete damaged plasticity model and laws of the anisotropy to model the structure. The compressive and flexure strengths of the studied microstructure loaded in three directions are compared with the experimental test results, as presented in Fig. 23. During these tests, no sliding and rotation from these specimens was observed. These criteria are assured to demonstrate the good conditions of the loading process. This Figure shows the simulation results, which are consistent with the anisotropic trend of the experimental results. They also validate the accuracy of the finite element model. For microstructures loaded in the x-direction, the flexural strength has the lower values, while microstructures loaded in the y- and z-directions have higher values. The compressive strength is the smallest when loaded in the z-direction. The compressive strength of the specimens has the most significant values, followed by the strength in the x- and y-direction. The experimental results are slightly higher than the numerical ones for the compression and flexural strength of specimens loaded in all directions except the compression one following the longitudinal direction.

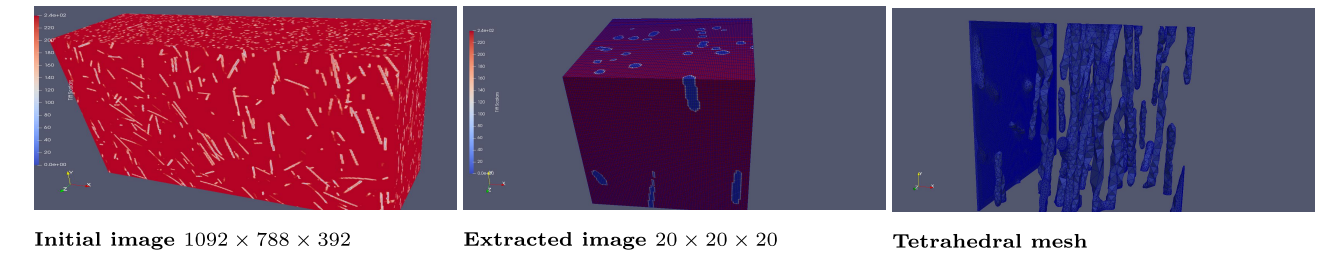


Figure 20. Extraction image and treatment mesh for the structure of hemp fibers.

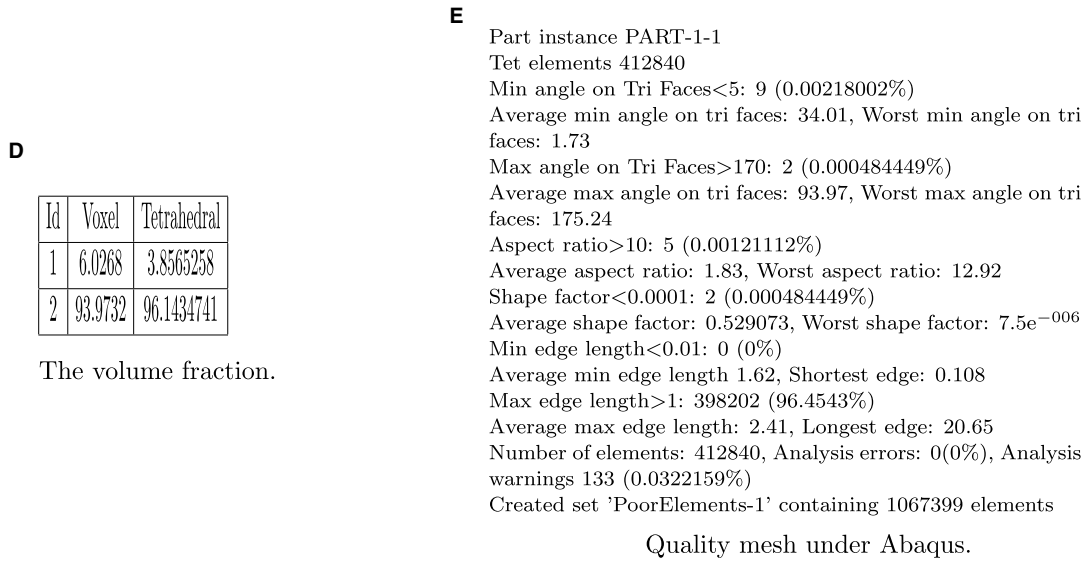
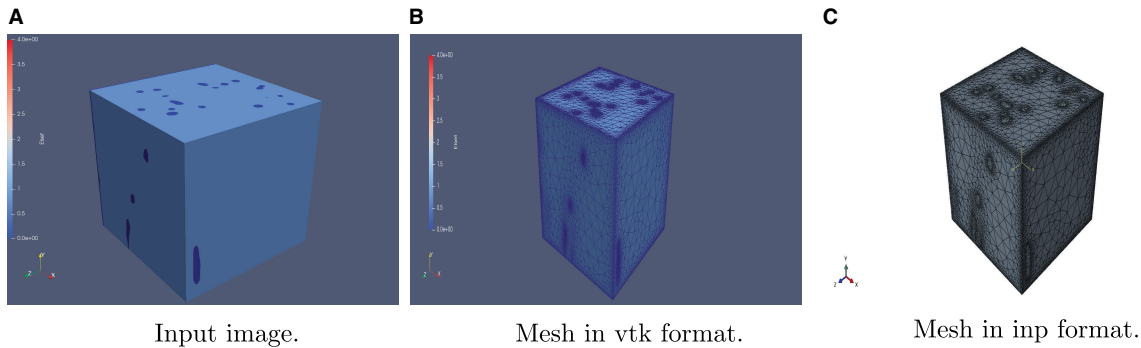


Figure 21. Generating conformal mesh and quality mesh for composite material microstructure with multiple material marching cubes algorithm (M3C).

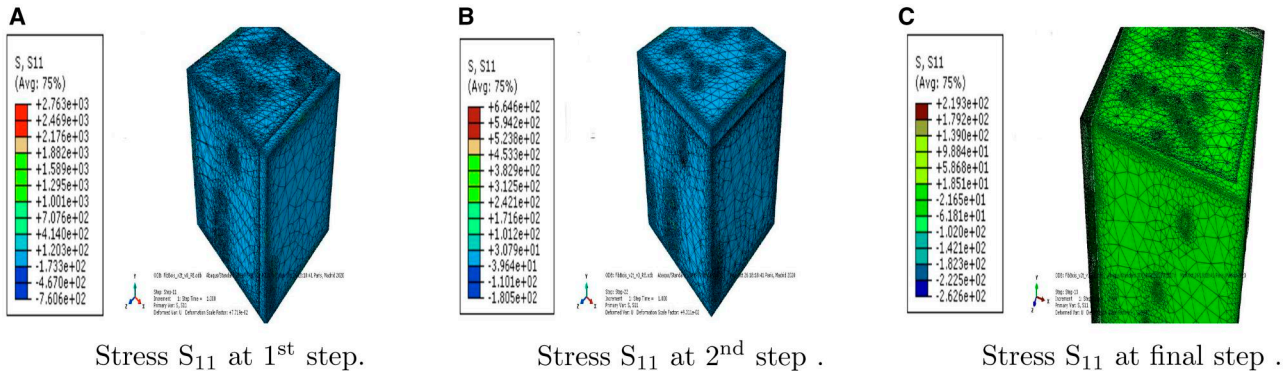


Figure 22. First component of stress S_{11} for the structure of hemp fibers under Abaqus over some steps.

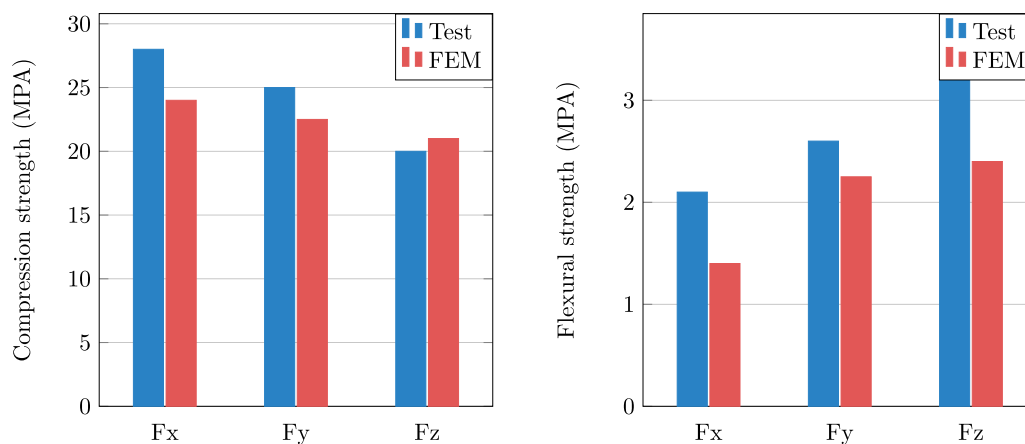


Figure 23. Comparison between FE and experimental test results [31] of some 3D microstructures under compression and flexure.

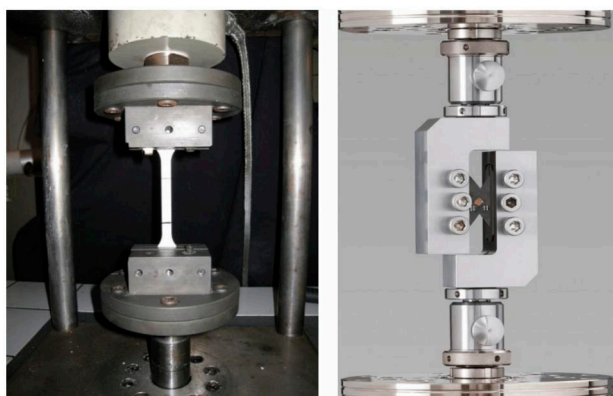


Figure 24. 3D Real view of the universal testing machine; the designed device is 15 cm wide, 55 cm long and 45 cm high as total dimensions.

The universal testing machine (UTM), also known as a universal tester, incorporated in this test belongs to ZwickRoell Company. Its capacity is 5000 kN in compression and tension. The standard ASTM was used to calibrate this machine with the suitable temperature. In this work, the design and calibration measurement of the universal testing machine for tension tests of materials is presented in Fig. 24.

To investigate the mechanical properties and stress-strain relationship model for the studied microstructures, many experiments have been carried out. We analyze these test results to provide the appropriate conclusions. Note that testing of different microstructure material needs to establish new standards and fix the test conditions, size and, geometry of the samples, to obtain reproducible mechanical properties. This machine provides tensile testing, which is an essential way to investigate one of these mechanical properties of materials, and it is a good technique for bulk characterization. The tensile testing was carried out at a constant strain rate at room temperature. Many samples used in this study of cells' representative volume element (RVE) geometry and dimensions are illustrated in the previous section. The reader is referred to the sheet of these samples for extra information about the physical and mechanical characteristics. During these tests, the load and machine movements are recorded. The test is performed using a microcomputer servo universal testing machine and a data acquisition system. The camera captures the front view of the cell. The mechanical test is interrupted when the fracture occurs. Each test configuration

was repeated at least twice to guarantee the correctness of these tests. Mechanical tests, including time and measurements, are performed to evaluate the physical properties, such as the modulus of elasticity according to the ASTM D-882-91 [87, 88]. Compressive strength is tested based on the ASTM standard [89, 90] on these specimens with various diameters and heights.

The load-displacement curves are illustrated in Fig. 25. The micrograph indicates that the phenomenon is shared into two stages. In the initial stage, a little noise can be heard around an important ultimate load which can cause small cracks in the zones of the specimens around this loading. The final stage consists of splitting the specimen into two main parts. The load increases steadily with respect to displacement. Note that there is no buckling during the loading.

The present study explores tensile deformation through experimental visualization and theoretical analyses. Experimental analyses reveal that the constitutive relation of tension behavior is characterized by three stages. Once an external tensile loading is applied to a specimen, elastic and plastic deformations are expected. First, an elastic deformation comes to the specimen, which results in a short linear relationship of an extension. The original and final length of the specimen and the external axial tensile load are needed to compute the engineering stress and strain. This yields the stress-strain proportionality as illustrated in Fig. 26. The elastic deformation is very short, which is followed by two different slopes, meaning uniform and non-uniform plastic deformation. During the uniform stage, the stress rises for permanent deformation from the continuous load. At the end of this stage, the stress reaches the highest value, meaning that the elastic limit occurs. Then, a decreasing stress occurs accordingly until fracture. The nonlinear deformation phenomenon is detected when this peak stress is attained. The stress-strain graph exploited in this study shows a domination of plastic deformation. The FEM framework results presented in this Figure show that it is a good agreement compared to the experimental observations for providing deformation and mechanical responses.

Summary and future work

We have presented a new technique for generating adaptive tetrahedral mesh from segmented 3D imaging data. The aim was to obtain a smooth material interface based on the marching cubes algorithm. The presented approach is much more acceptable for managing complex microstructure geometry. Raising the number

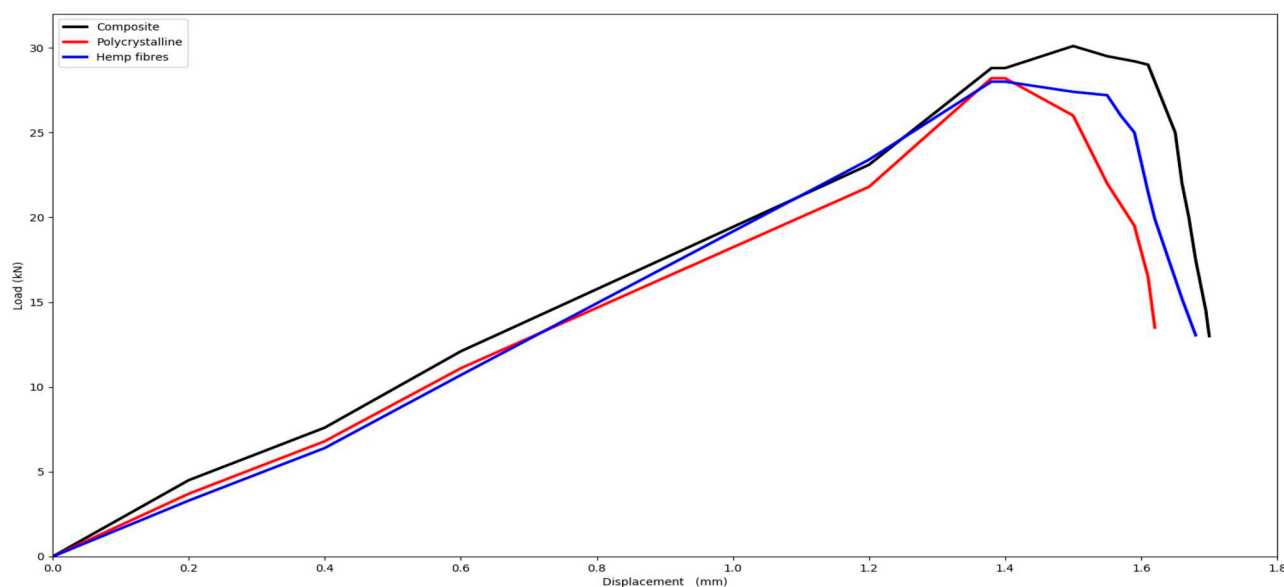


Figure 25. Experimentally obtained load-displacement for shear specimens.

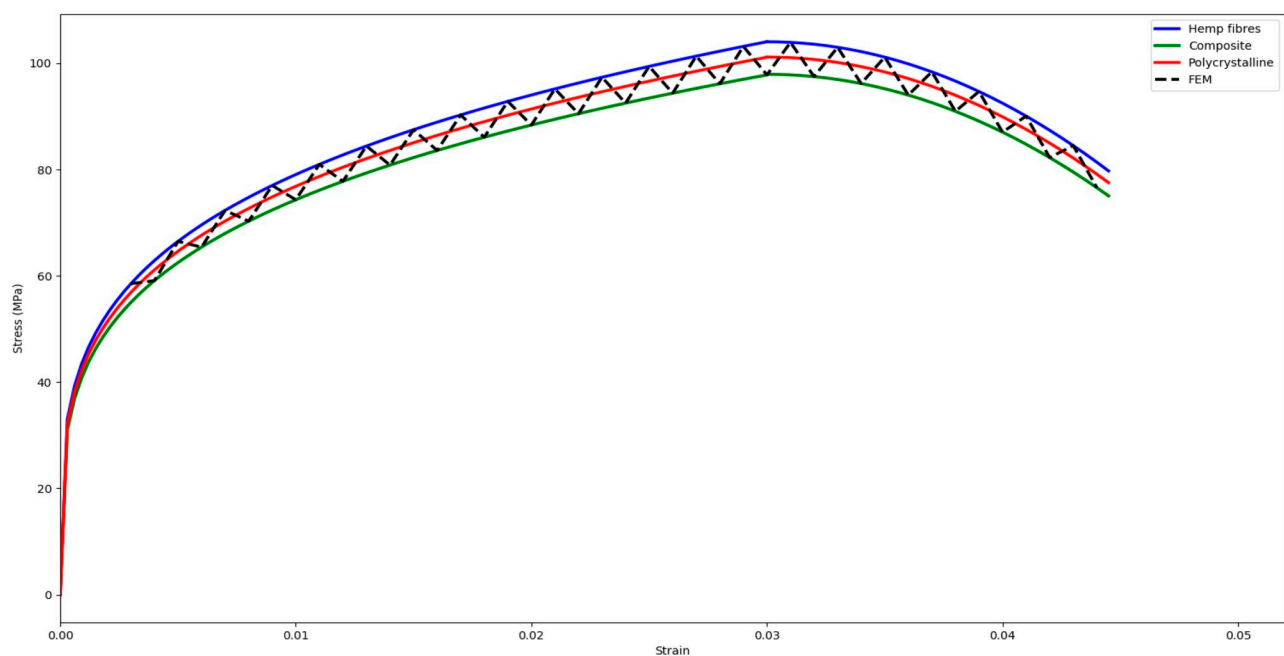


Figure 26. Stress-strain curves obtained from the universal testing machine, showing the elastic and plastic behavior of material.

of remeshing iterations is recommended. This smoothing method leads to minimizing the surface area of the microstructure. We have shown that generating a good-quality mesh is necessary to establish an adequate FEM model with promising robust simulations.

In this study, we have suggested mesh generation from different microstructure images. All Abaqus post-processing modules are created, material characteristics, are defined and explained, achieving their capability to be functional. Our concrete work is summarized with getting a realistic model accompanied by performed simulations. In future work, we would like to compute the error estimation from adaptive mesh refinement. The primary objective of this study could be achieved by comparing the

quality mesh obtained by the solver and some available tools such as Paraview. Further, the enrichment will be presented by implementing the eight-node hexahedral elements (FE-H8) in place of the four-node tetrahedral elements (FE-T4) applied to linear and non-linear elasticity problems for good performance. The aimed model has been shown in previous works [91, 92] to generate robust results and better accuracy than the standard FE.

In this paper, we have started from μ -CT data of materials to generate a realistic FE model for different microstructure images. We have presented these results in the Abaqus solver to show the ability of the framework to forecast the mechanical properties of any given material microstructures. Moreover, this paper

investigated the classical homogenization method results in terms of effective material properties depending upon the inclusion volume fraction. Future work will focus on simulation and homogenization aspects applied to various microstructural constituents. The aim is to interpret profoundly realistic deformation paths with simulation and homogenization aspects. This improvement contributes to building viscoelastic modeling, including anisotropy behavior, on both the microscopic and macroscopic levels since there are mechanical limitations in the existing model.

Numerous sophisticated software tools with parallel architectures have been developed in various scientific domains with satisfactory performance. One of them is a PETSc library [93], which is maintained by Argonne National Laboratory, and is organized with data structures and routines for scientific computing. The future contribution is aimed at providing a new computation chain based on this library. This excellent parallel efficiency is expected to be better than the Abaqus solver.

The future investigation will aim to study the stress-strain behavior at different temperatures systematically. For the purposes of this modeling, a means of obtaining stress-strain with dependency on the temperature is required. Specimens will be machined in the through-thickness direction over various ranges of temperature. Following this strategy, we will present the results of the compression and tension testing as a function of temperature. It is interesting to analyze the stress-strain response concerning a temperature variation. Investigating the tensile test data versus temperature and strain rate provides additional information about these properties and the fracture mechanisms. The thermo-mechanical process is crucial for determining critical points of the hot stress-strain curve. In this investigation, this modeling permits us to illustrate the behavior of material during hot deformation, such as hot rolling. Determining the critical strain on this curve has been described in many works [94–96]. A transition in the deformation mode will be observed, particularly the plastic one that occurs at sensitive temperatures.

We have shown via the stress-strain curve that the different investigated materials possess good mechanical properties, exhibiting high true stress (100 MPa) and considerable true strain (0.031) with relatively low yield strength and continuous yielding. Future work can present the true strain in terms of the volume fractions of a phase constituting each material. The objective is to determine the true strain ranging zones where it is invariant from the volume fraction. This research investigates the deformation mechanisms by correlating the stress-strain curve. For this, we would like to present the derivation of the true σ - ϵ curve after smooth treatment using the Abaqus solver. Then, the graph of strain hardening rate $\theta = d\sigma/d\epsilon$ will be analyzed to provide the corresponding different deformation. During this deformation process, more deformation twins could be formed. Some micro-cracks that grow along the interface of different phases can be created when failure occurs. The variations of work-hardening parameters and θ - σ as a function of temperature and strain rate can be investigated to determine distinct temperature regimes.

The textile composite material is introduced in Section “Fiber orientations with segmentation aspect” with multiple layer orientations. Textile composite material at mesoscale remains a challenging stage of material modeling. The objective is to continue performing a mesoscopic analysis of textile reinforcement deformations. For this, the fiber direction should be determined at each point of the textile material, because the textile material analyzed in this study exhibit three directions of anisotropy. Using this material parameter facilitates taking into account the

variability of the reinforcement that generally decreases its rigidities. These orientations (Equation 1) of this material are computed thanks to the calculation of the structure tensor. Note that the quality of the result of the meso FE modeling actively depends on the FE mesh. Our accurate and advanced meshing algorithm is used to avoid such poor quality. Another perspective on the work is to run a framework using computed tomography to determine FE models based on the real geometry of the textile reinforcement. The mechanical properties will be identified from tensile tests. Spatial distribution of fibers for different deformations could be performed with experimental observations and compared to the FE model.

Data availability

Data not available due to ethical and commercial restrictions.

Author's contributions

Tarik Chakkour Conceptualization; Methodology; Validation; Formal analysis; Investigation; Data Curation Management; Writing-Original Draft; Visualization.

Conflict of interest statement: We have no conflicts of interest to disclose. All authors declare that they have no conflicts of interest.

Data availability

Data will be made available on request.

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Consent to participate

All authors contributed to this work. All authors read and approved the final manuscript.

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